



Research review paper

A journey with psychedelic mushrooms: From historical relevance to biology, cultivation, medicinal uses, biotechnology, and beyond

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ABSTRACT

Psychedelic mushrooms containing psilocybin and related tryptamines have long been used for ethnomycological purposes, but emerging evidence points to the potential therapeutic value of these mushrooms to address modern neurological, psychiatric health, and related disorders. As a result, psilocybin containing mushrooms represent a re-emerging frontier for mycological, biochemical, neuroscience, and pharmacology research. This work presents crucial information related to traditional use of psychedelic mushrooms, as well as research trends and knowledge gaps related to their diversity and distribution, technologies for quantification of tryptamines and other tryptophan-derived metabolites, as well as biosynthetic mechanisms for their production within mushrooms. In addition, we explore the current state of knowledge for how psilocybin and related tryptamines are metabolized in humans and their pharmacological effects, including beneficial and hazardous human health implications. Finally, we describe opportunities and challenges for investigating the cultural production of psychedelic mushrooms and metabolic engineering approaches to alter secondary metabolite production through biotechnology approaches integrated with machine learning. Ultimately, this critical review of all aspects related to psychedelic mushrooms represents a roadmap for future research efforts that will pave the way to new applications and refined protocols.

1. Introduction

Despite the long history of traditional use for spiritual and medicinal purposes, the potential importance of psychedelic mushrooms in modern society has not been fully explored. The psychoactive effect of these mushrooms is commonly associated with the tryptamines psilocybin and psilocin. Psilocybin was first isolated by Albert Hofmann in the late 1950s (Hofmann et al., 1959) and then marketed for experimental and therapeutic use throughout the 1960s (Bogadi and Kaštelan, 2021; Geiger et al., 2018; Passie et al., 2002). However, in 1971, an international framework was imposed by the United Nations (Stebelska, 2016) identifying psilocybin as an addictive substance void of medicinal value (Fricke et al., 2019; Plotnik et al., 2022). Yet as evidence of therapeutic potential associated with psychedelic mushrooms has emerged, many countries have begun to step away from this framework (Plotnik et al., 2022). For example, psilocybin-producing mushrooms have potential therapeutic value for treating various forms of depression (Carhart-

Harris et al., 2017; Fraga et al., 2021; Nkadameng et al., 2020), obsessive compulsive disorder (Lugo-Radillo and Cortes-Lopez, 2021; Moreno et al., 2006; Wilcox, 2014), anxiety (Bogadi and Kaštelan, 2021; Grob et al., 2011), and addiction (Bogenschutz and Pommy, 2012; Johnson et al., 2014). Since early 2022, the Canadian government has allowed physicians to prescribe psilocybin for emergency treatment measures through Canada's Special Access Program (Government of Canada, 2022) and the U.S. state of Oregon has legalized psilocybin and made it available to adults (Oregon health Authority, 2022). Given these recent policy changes, it is expected that additional avenues for medicinal uses of psychedelic mushrooms and psychedelics in general will become more readily available in various parts of the world.

There are many different types of psychoactive mushrooms which do not contain psilocybin, but produce various other psychoactive compounds. However, mushrooms containing psilocybin and related compounds, with the most commonly available sources being *Psilocybe*

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mushrooms, remain the focal point of this review. Herein, we discuss traditional relevance and use of *Psilocybe* mushrooms, their taxonomy and geographical distribution, as well as trends in psilocybin research. With respect to the latter, the review includes information on extraction and quantification, pharmacokinetics and pharmacodynamics of magic mushroom secondary metabolites, health and hazardous implications, cultivation practices, and current/future opportunities based on biotechnological advances and machine learning related primarily to *Psilocybe* mushrooms.

1.1. Traditional significance of psychedelic mushrooms for indigenous cultures

Although psychedelic mushrooms are only beginning to be re-evaluated as therapeutic treatment options by western medicine, humans have a much deeper history and relationship with psychoactive mushrooms such as *Psilocybe* spp. The use of psychoactive mushrooms may have occurred with our *Homo sapien* ancestors, including hominids of the Pliocene epoch (Rodríguez Arce and Winkelman, 2021). Relatively more recent use is implied by the discovery of prehistoric 6000 to 8000 year old rock drawings, respectively found in Spain and the Saharan mountains, that depict that ancient peoples had knowledge of the psychoactive effects of psychedelic mushrooms (Samorini, 2019). Moreover, psychoactive mushrooms were used in spiritual rituals by numerous ancient civilizations as far back as 1500 BCE (Carod-Artal, 2015). There exists evidence supporting the regular use, collection, and/or cultivation of psychedelic mushrooms by different Indigenous civilizations across the globe throughout ancient history. These psychoactive mushroom species have been postulated to serve as important resources for spiritualistic and communal gatherings within early Indigenous societies (Rodríguez Arce and Winkelman, 2021). Additional and more detailed examples of the historical relevance and value of psychedelic mushrooms to different ancient and Indigenous cultures are highlighted below.

The ancient Chinese doctrine “Baopuzi” (translated as Master Who Embraces Simplicity) written by Daoist philosopher and alchemist Ge Hong (283 to 343 A.D) describes many intoxicating mushrooms. These include descriptions of *Psilocybe* mushrooms, among others, as proposed agents for achieving immortality (Chen, 2021). Ge Hong described many “zhi” (translated as Spirit Mushrooms) mushrooms, including “Qingyun zhi” (Black Cloud Spirit Mushroom), “Mamu zhi” (Female *Cannabis* Spirit Mushroom), “Rou zhi” (Flesh Spirit Mushroom), “Hu zhi” (Tiger Spirit Mushroom), “Mingniao zhi” (Chirping Bird Spirit Mushroom), among others that “...grow on famous mountains. Eating them will enable one to ride on clouds and view the Eight Extremities and communicate with the spirits.” (Chen, 2021). Modern interpretations of Ge Hong's work explain the importance of mind-altering substances, including psychedelic mushrooms, in relation to certain areas of ancient Chinese philosophy.

There is also a history of use of psychoactive mushrooms in India. Several different species of *Psilocybe* (Thomas, 2002) and *Panaeolus* (Kaur et al., 2014) that are known to produce psilocybin are native to India. The Khajuraho Temples of India, built between 900 and 1130 CE, house erotic sculptures depicting orgies among humans with disjointed limbs in anatomically impossible positions, which are thought to represent mystical shamanic dismemberment experiences (Maillart-Garg and Winkelman, 2019). There also exist many ancient Khajuraho sculptures that depict mushrooms. The natives living in this region of India possess ethnomycological knowledge relating to a diverse array of psychedelic mushroom species (Winkelman et al., 2022). Maillart-Garg and Winkelman (2019) propose that the central placement of mushroom sculptures represent their association with Vedic, Jain, Hindu, and/or Buddhist gods. Thus, it is possible that psychoactive mushrooms were a significant part of certain ancient South Asian religious practices and Vedic texts.

Indigenous Mesoamerican cultures shared immersive connections with *Psilocybe* species before the arrival of the Spanish in the 16th century. It is well documented that Indigenous communities of this historical region considered *Psilocybe* mushrooms to have both medicinal and sacred significance for use in both religious rituals and in daily life (Guzmán, 2008; Lowy, 1980). Although traditions of societies such as the Mazatec, Aztec, and Zapotec were oppressed with colonization, representations of their beliefs and rituals persist within different pieces of art, ceramics, stone, and various codices (Guzmán, 2003; Lowy, 1980). For example, Aztec codices are manuscripts written and illustrated in both Spanish and their native Nahuatl in attempts to preserve the Aztec civilization through the colonial period of what has since become Mexico (Illana Esteban, 2013). These codices describe daily life through illustrations that depict native flora, fauna, customs, and ceremonies. The 16th century Codex Magliabechiano contains a portrayal of a human ingesting “Teonanácatl” (translated as Flesh of the Gods), thought to be *Psilocybe mexicana*, while being followed by a creature that represents the mystical properties of *Psilocybe* spp. (Guzmán, 2016; Schultes, 1940).

The psychoactive mushroom taxon was first incorrectly identified by Reko (1919) and by Schultes (1940) then later described as “Teonanácatl” in “Historia de las cosas de Nueva España” by Fray Bernardino de Sahagún, who wrote “The little mushrooms that grow in this land are named ‘Teonanácatl’. They grow beneath the hay in the fields and plains. They are round, and their stems are tall and round and slender. Their taste is unpleasant; they cause sore throat and drunkenness. They are used as medicine for fever and gout. No more than two or three should be eaten. Those who eat them see visions and feel fluttering of the heart; the visions they see are sometimes frightening and sometimes humorous. Those few who eat them in excess are driven to lust. Silly and naughty boys are told that they have eaten ‘Teonanácatl’” (Carod-Artal, 2015). These accounts represent early works attempting to attribute traditional knowledge to modern taxonomy.

1.2. Diversity and distribution

Psilocybe spp. have a broad global distribution. Wide distribution of mushroom genera can result in ecotype-like adaptations which can broaden genetic diversity and associated phenotypes (Frings et al., 2020). Taxonomic arrangement of *Psilocybe* mushrooms has been widely debated and re-worked based on continuously emerging data (Cortés-Pérez et al., 2021). Macro- and micro-morphological features (Guzmán, 2012; Guzman et al., 2014; Rossato et al., 2009) which were previously used to describe taxonomy among psychedelic mushrooms are largely inefficient to describe the variations between the wide range of known psychedelic mushroom species. Thus, current methods in the taxonomical ranking of psychedelic mushrooms relies on sequencing the Internal Transcribed Spacer (ITS) region of the ribosomal operon, representing a barcoding region used to sufficiently differentiate between species (Gotvaldová et al., 2022).

Several genera of psychedelic mushrooms containing psilocybin have been documented and include *Psilocybe*, *Panaeolus*/*Copelandia* (Carod-Artal, 2015; Gotvaldová et al., 2022), *Inocybe* (Gotvaldová et al., 2022; Wieczorek et al., 2015), *Pholiotina*, *Pluteus*, *Galerina*, and *Gymnopilus* (Gotvaldová et al., 2022) which are largely polyphyletic. Modern attention is largely focused on the genus *Psilocybe* (Fr.) P. Kumm., which is globally distributed (Carod-Artal, 2015; Guzmán, 2005). Discoveries of novel species and subspecies continue to be described (Borovička et al., 2012; Li et al., 2014; Ma et al., 2014; Ma et al., 2016).

In 2007, Redhead et al. (2007) submitted a proposal to officially differentiate these mushrooms into separate genera, which was accepted by the Nomenclature Committee for Fungi (Norvell, 2010). This emendation broadly divided the previous *Psilocybe* into blue-bruising, neurotropic species to be categorized within the *Psilocybe* sensu stricto,

whereas mushroom types lacking such features were re-classified in the genus *Deconica* (W.G. Sm.) P. Karst. (Guzmán, 2012; Ramírez-Cruz et al., 2013; Rossato et al., 2009). This modification retained both genera as members of the Strophariaceae. More recently, molecular-based computational approaches combined with a micromorphological component have re-created the phylogeny of *Psilocybe* mushrooms (Ramírez-Cruz et al., 2013). The aforementioned study based their analysis on three nuclear markers, indicating *Deconica* and *Psilocybe* to be respective members of Strophariaceae and Hymenogastraceae, each with separate monophyletic family ties. Additionally, ancestral state reconstruction determined that common micromorphological characteristics previously used to identify *Psilocybe* and *Deconica* were homoplastic, evolving in isolated instances tied to their independent lineages, and suggesting that these traits are not reliable for taxonomic classification (Ramírez-Cruz et al., 2013). Although this re-classification of *Psilocybe* among the Hymenogastraceae is generally accepted (Cowan and Elkins, 2018; He et al., 2019), some researchers continue associating *Psilocybe* among the Strophariaceae (Guzmán et al., 2014).

The concept of species, as it pertains to psychedelic mushrooms, is constantly changing. Emergent species and the re-characterization of species to reflect new data is continuous process. Cortés-Pérez et al. (2021) assessed twenty-eight type collections of *Psilocybe* mushrooms. When these collections were re-assessed, only five species were recognized (Cortés-Pérez et al., 2021). Additionally, Borovička et al. (2011) suggest categorizing select taxa related to *Psilocybe serbica* as various varieties and forma of *P. serbica*, in addition to re-defining a sectoid fungus *Weraroa novae-zelandiae* as *Psilocybe weraroa* (Borovička et al., 2011). Thus, additional, continuous research must be completed to better determine the phylogenetic nature of psychedelic mushrooms.

There exists evidence supporting horizontal gene transfer between psilocybin producing *Psilocybe* and *Panaeolus* (Reynolds et al., 2018). Some psilocybin biosynthesis enzymes and intermediates are undetectable in certain species of *Massospora*, which are entomopathogenic fungi that produce psilocybin, suggesting a potentially novel psilocybin biosynthesis pathway (Boyce et al., 2019). McTaggart et al. (2023) found that *Psilocybe subaeruginosa* undergoes tetrapolar mating, and also possesses three copies of the *psiH* gene, which encodes for tryptamine monooxygenase, the second step in the psilocybin biosynthesis pathway. Other *Psilocybe* species, including *P. azurescens* and *P. cyanescens* also have three *psiH* homologs (McTaggart et al., 2023). Although *P. cubensis* is a coprophilous species (Quinde and Niveiro, 2021), *P. subaeruginosa*, *P. azurescens* and *P. cyanescens* represent wood-degrading species that might be synonymous with one another based on unpublished data. The functional relevance of each of the three *psiH* homologs remains to be determined (McTaggart et al., 2023). Ultimately, it has been suggested that psilocybin synthesis offers a fitness advantage in certain coprophilous and saprotrophic fungal niches, but the precise benefit is not fully understood (Reynolds et al., 2018). However, in this regard, several hypotheses have been suggested.

Although there is little evidence, Lenz et al. (2021) present various hypotheses to suggest ecological benefit of psilocybin production. The “monomer hypothesis” proposes that psilocybin-producing mushrooms neurochemically manipulate the behaviour of animals for protection, or in the case of *Massospora*, to aid in host infection. The “polymer hypothesis” describes that oligomerization of psilocin triggers a blueing reaction due to the formation of to psilocyl radical derivatives that that might react with proteins to negatively impact insect feeding. Finally, the production of β -carboline by *Psilocybe* mushrooms can act to prevent inactivation and elimination of bio-active psilocin following psilocybin ingestion. It is suggested that β -carboline act synergistically with psilocin to intensify psychoactive effects (Lenz et al., 2021), perhaps to extend the benefits of the “monomer hypothesis” to larger animals.

1.3. Scientific records and research trends

Although the traditional significance of psychedelic mushrooms dates back thousands of years (Rodríguez Arce and Winkelmann, 2021), scientific investigations of *P. cubensis* are quite recent, originating in the 20th century (Fig. 1). This section presents an audit of the scientific literature related to psychedelic mushrooms while describing different research areas related to distinct fields of study. Ultimately, trends in related research interest and the volume of scientific papers being produced in relation to each era are depicted.

Magic mushroom-related publications were searched across four major scientific literature and citation databases for biomedical and life science journals, specifically Web of Science of Clarivate Analytics (Web of Science of Clarivate Analytics, 2023), PubMed Central at NCBI (PubMed Central at NCBI, 2023), Elsevier's Scopus (Elsevier's Scopus, 2023), and EuropePMC (EuropePMC, 2023). “Magic mushroom OR *Psilocybe* OR psilocybin OR psilocin” were used as the search criteria to examine available scientific publications. Nearly 3429, 2739, 2280, and 525 magic mushroom-related scientific papers were respectively found in EuropePMC, NCBI PMC, Web of Science, and Scopus.

Based on the psychoactive mushroom-related scientific publications between 1925 and 2022, we divide the scientific era into three periods (Fig. 1, A). In Period I (1925 to 1959), mainly reported taxonomical aspects, cultivation methods of psychedelic mushrooms, and their application in traditional medicines. In the early part of Period II (1959 to 2017), research described the medicinal potential and chemical properties of psychedelic mushrooms; thereafter studies elucidated the psilocybin biosynthesis pathway. Although scientific efforts to understand and uncover the medicinal applications of psychedelic mushrooms are not yet fully recognized, period II is associated with a series of advancements in the direction of the medicinal potential and chemical properties of psychedelic mushrooms, including the isolation, and synthesis of psilocybin and psilocin. Period III (2017 to 2022) is more related to biotechnology and metabolic engineering of *Psilocybe* spp. and shows unparalleled growth in studying psychedelic mushrooms. The production of psilocybin in heterologous systems was the main accomplishment that significantly accelerated progress in this field. Trends related to the study of psychoactive mushrooms demonstrates the exponential increase of interest within the scientific community (Fig. 1, A).

For further analysis, the published papers were classified into seven groups (Fig. 1, B) including (i) pharmaceutical (i.e., pharmaceuticals, pharmacology, toxicology, and drug), (ii) medicinal (all medicinal and medical topics), (iii) biochemistry (i.e., biochemistry, virology, microbiology, biotechnology, and chemical studies), (iv) neurosciences (i.e., psychology, psychiatry, brain-related, neurological, and cognitive studies), (v) bioinformatics (i.e., computational biology and data analytics), (vi) mycology (i.e., fungi science, cultivation, taxonomic aspects, molecular biology, cell biology), and (vii) social (i.e., law, social, and policy-based studies). The diversity of disciplines presented in this analysis shows various evolving areas of interest. Neurosciences (42%) and pharmaceutical aspects (23%) of *Psilocybe* spp. have attracted the most attention, followed by biochemistry (14%) (Fig. 1, B). In contrast, mycology and cultivation comprised only 10% of the published scientific papers, followed by 7% medicinal aspects, 3.4% of social subjects, and 0.6% of bioinformatics-based psychedelic mushroom research (Fig. 1, B).

1.4. Biosynthesis, detection, and quantification of secondary metabolites in magic mushrooms

Due to their psychoactive potential, psilocybin and its dephosphorylated counterpart psilocin are generally of higher interest and are thus more commonly tested for. The psychotomimetic potency of psychoactive mushrooms is generally attributed to the species or variety in question (Gotvaldová et al., 2022). However, there exists vari-

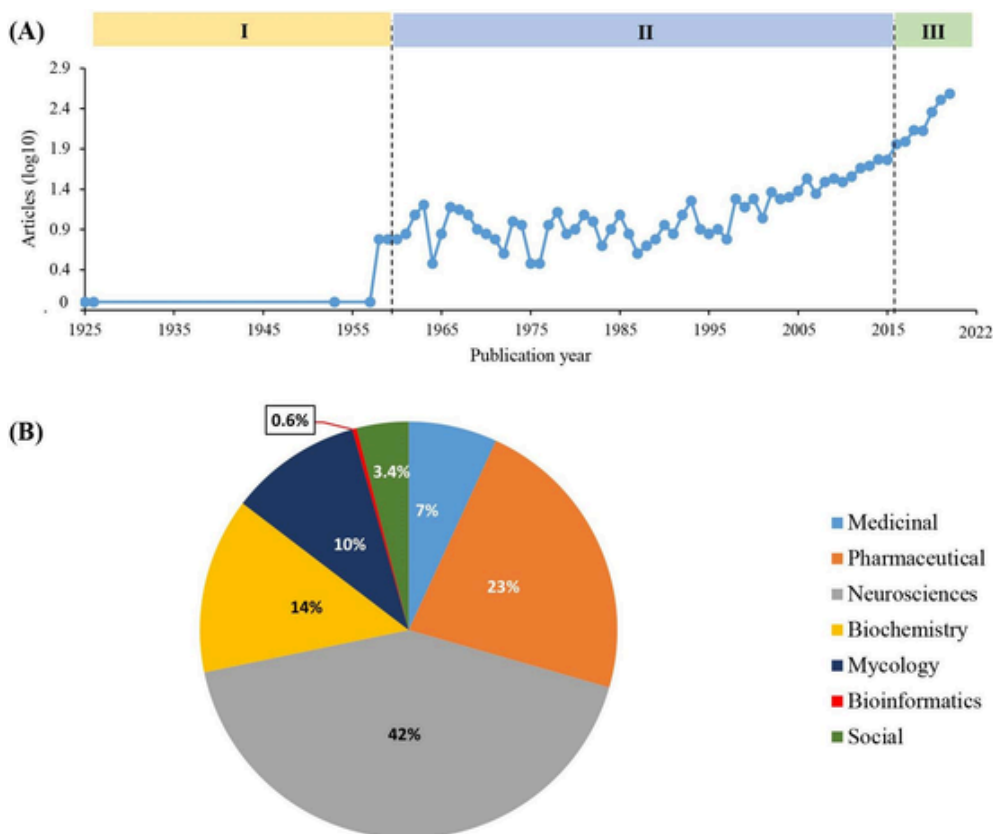


Fig. 1. Research trends of *Psilocybe* sp. (A) The overall pattern of *Psilocybe* research from 1925 to 2022 is displayed by the trend line (log scaled) with each dot depicting the number of scientific papers (Y-axis) per year (X-axis). The scientific research periods are classified as I (1925 to 1959), II (1959 to 2017), and III (2017 to 2022) and are split with vertical dotted lines. (B) The classification of *Psilocybe* research subjects including social, medicinal, pharmaceutical, neuroscience, biochemistry, mycology, and bioinformatics.

ability within those varieties and differences based on cultivation and harvesting strategies, which will be discussed later. Although psilocybin and psilocin are the primary tryptamines commonly used for testing psychedelic mushrooms, there are additional tryptamines and derivative compounds of high interest to various research fields. Thus, the following sub-sections outline the current state of knowledge on biochemical mechanisms underlying biosynthesis of secondary metabolites, their detection and quantification.

2. Biosynthesis of secondary metabolites in psychedelic mushrooms

In *P. cubensis* and *P. cyanescens*, tryptamine biosynthesis from the amino acid L-tryptophan, culminating in the production of psilocybin, involves decarboxylation, hydroxylation, phosphorylation, and methylation reactions (Jensen, 2004; Lenz et al., 2021) (Fig. 2). Fricke et al. (2017) report a locus containing psilocybin biosynthesis genes (*psiD*, *psiH*, *psiK* and *psiM*) in *P. cubensis* and a syntenic region in *P. cyanescens*. The recent release of a draft genome for the *P. cubensis* strain “Penis Envy” has identified a contig containing psilocybin pathway genes (McKernan et al., 2021b). The process begins with decarboxylation of L-tryptophan by tryptophan decarboxylase (PsiD) to tryptamine, followed by P450 monooxygenase (PsiH)-mediated hydroxylation to 4-hydroxytryptamine. Thereafter, the psilocybin intermediates norbaeocystin and baeocystin are formed with sequential phosphorylation and methylation by 4-hydroxytryptamine kinase (PsiK), and psilocybin synthase (PsiM), respectively. Norpsilocin is a dephosphorylated derivative of baeocystin, and this byproduct does not appear to be involved in psilocybin synthesis (Fig. 2) (Lenz et al., 2021); the phosphatase involved in norpsilocin formation is unknown. PsiM also catalyzes con-

version of baeocystin to psilocybin using S-adenosylmethionine (SAM) as a methyl donor (Fig. 2) (Fricke et al., 2017). (Blei et al., 2020) discovered TrpM, an alternative SAM-dependent methyltransferase found in *P. serbica*, which catalyzes a distinct, PsiM-independent pathway for mono- and di-methylation of L-tryptophan, which is unrelated to psilocybin biosynthesis. Although chemical synthesis of aeruginascin from psilocybin has been achieved in vitro (Sherwood et al., 2020), no similar in vivo reaction has been described. Interestingly, the recombinant form of PsiK from *P. cubensis* and *P. cyanescens* can respectively phosphorylate psilocin and 4-hydroxy-*N,N,N*-trimethyltryptamine (4-HO-TMT) to psilocybin and aeruginascin (Dörner et al., 2022). The occurrence of 4-HO-TMT in psychedelic mushrooms and in vivo PsiK mediated formation of aeruginascin have not been described in the literature.

Some mushrooms become blue when damaged, which is described as “bruising”. Although not all mushrooms that bruise are psychedelic mushrooms, it is common for *Psilocybe* spp. to become blue-green color when damaged. The blue color of damaged psychedelic mushrooms appears to be associated with the accumulation of quinoid polymers resulting from the successive dephosphorylation of psilocybin to psilocin by PsiP and oxidation by an O_2 -dependent laccase (PsiL) to yield quinoids which then polymerize to form various oligomers (Lenz et al., 2020). To date, the enzymatic steps driving this polymerization are unknown and may occur non-enzymatically following enzyme activity. It is worth mentioning that PsiP purification from *P. cubensis* mushrooms was followed by selecting for phosphatase activity, tested with the synthetic substrate *p*-nitrophenyl phosphate (Lenz et al., 2020). Although the aforementioned final PsiP preparation converts psilocybin to psilocin, its substrate preference is unknown and thus the possibility remains that PsiP is a non-specific phosphatase; similar non-specific phos-

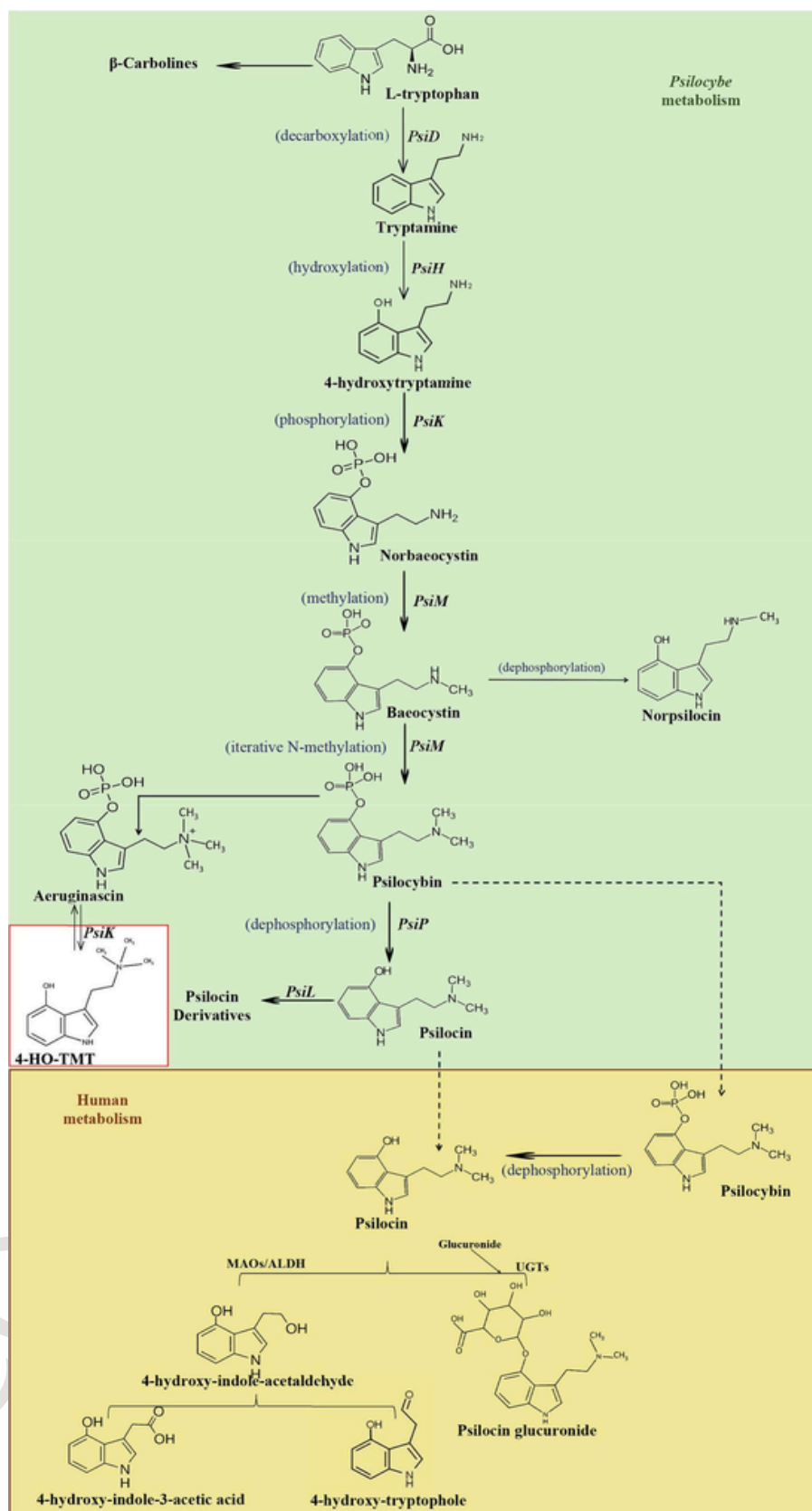


Fig. 2. A comparison of tryptamine biosynthesis in *Psilocybe* mushrooms (green) and metabolism of psilocybin and psilocin in humans (beige). Undefined processes are also outlined (red). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

phatases have been described in the literature (Milne et al., 2020). Therefore, an additional psilocybin phosphatase may exist in psychedelic mushrooms. To date, not much is known with respect to the precise structural steps that convert L-tryptophan to β -carbolines in psychedelic mushrooms. β -carbolines are further described in later sections.

3. Detection and quantification of secondary metabolites

Psychedelic mushrooms are the source of the well-known psilocybin and other tryptamines (e.g., baecystin, aeruginascin, and certain dephosphorylated derivatives that in some cases, can also accumulate in these mushrooms). On a dry weight basis, psilocybin content within the basidiocarp ranges from 0.03% to 1.6% across various *Psilocybe* species (Anastos et al., 2006; Beug and Bigwood, 1982; Fricke et al., 2019; Gartz et al., 1994; Pedersen-Bjergaard et al., 1997). However, newly developed lineages with higher average amounts of psilocybin are continuously being produced. A recent survey of 14 *Psilocybe* species sourced from around the world established that tryptamine content including psilocybin is greatest in *P. cyanescens*, whereas *Psilocybe fuscifulva* contains no psilocybin (Gotvaldová et al., 2022). Tryptamines can also vary across varieties of a species, and their individual collections as is the case for *Psilocybe serbica* varieties 'arcania', 'bohemia' and 'moravica' (Borovička et al., 2015; Gotvaldová et al., 2022; Reynolds et al., 2018). Psilocybin levels can be on par to an order of magnitude greater than psilocin levels (Borner and Brenneisen, 1987; Gartz et al., 1994; Gotvaldová et al., 2022). The lower ratios are presumably due to spontaneous dephosphorylation of psilocybin to psilocin when basidiocarps are dried at room temperature or higher, whereas higher detected ratios are evident in lyophilized mushrooms (Lenz et al., 2017). Alternatively, different ratios of psilocybin and psilocin may be due to decomposition of psilocin at room temperature (Nichols, 2020) and/or hydrolysis-mediated conversion of psilocybin to psilocin (Kargbo et al., 2022). With respect to other tryptamines, baecystin content is absent in some psilocybin-containing species but as high as 0.45% on a dry weight basis in certain species (Christiansen and Rasmussen, 1982; Gartz et al., 1994; Gotvaldová et al., 2022; Pedersen-Bjergaard et al., 1997). Finally, aeruginascin is also present in the fruiting bodies of various *Psilocybe* species, albeit liquid chromatography mass spectrometry (LC-MS) analysis has established that it can be present at levels that are three orders of magnitude lower than psilocybin (Dörner et al., 2022). This trimethylated tryptamine was first identified when high-performance liquid chromatography (HPLC) and thin-layer chromatography (TLC) were used to analyze extracts prepared from fruiting bodies of *Inocybe aeruginascens* (Gartz, 1989a, 1989b; Jensen et al., 2006). Although it is barely detectable in some psilocybin-producing mushrooms, aeruginascin can be as high as 0.30% of the dry weight of the basidiocarp, as is known for *Inocybe corydalina* (Gartz, 1989a, 1989b). Apart from the abovementioned tryptamines, β -carbolines such as harmaline and harmine are also known to occur in *Psilocybe*, albeit at 0.1% or less of psilocybin levels in these same mushrooms (Blei et al., 2020).

In addition to the postharvest drying methods used for basidiocarps, extraction and analysis methodologies can impact secondary metabolite profiling and quantification thereof. Typically, methanolic solutions are used to extract tryptamines including psilocybin and psilocin from *Psilocybe* before taking the extract to dryness and resuspension in an organic solvent prior to downstream metabolite analyses (Beug and Bigwood, 1982; Bigwood and Beug, 1982; Blei et al., 2020). Protective measures such as purging extracts with N_2 gas and/or wrapping the tubes holding these in aluminum foil can also be used to limit spontaneous breakdown of tryptamines (Gotvaldová et al., 2022). Extraction can be completed with ultrasonic-assisted extraction of methanolic mixtures (Wieczorek et al., 2015). By comparison, β -carbolines can be extracted with slightly acidified aqueous solutions containing methylene chloride (Blei et al., 2020). To date, there is little evidence of standard

protocols for quantifying secondary metabolites extracted from psychedelic fungal materials that are based on the recovery of an added metabolite. Daily analysis entails the mathematical monitoring of the percentage of spike of a metabolite or internal standard to that was added to a duplicate sample; such an approach is the standard for analytical chemistry. TLC and HPLC are well established methods of detecting secondary metabolites of *Psilocybe* mushrooms (Anastos et al., 2006; Beug and Bigwood, 1982; Bigwood and Beug, 1982; Gartz, 1989a, 1989b; Marcano et al., 1994). Although both methods are commonly used, HPLC operated in either normal or reverse-phase modes are preferred over TLC (Pedersen-Bjergaard et al., 1997), as the latter has low resolution and quantitative capabilities. Compounds of interest can also be detected using electrochemical, UV, or fluorescence detection methods (Christiansen and Rasmussen, 1983; Marcano et al., 1994; Saito et al., 2005). However, Anastos et al. (2006) found improved sensitivity for psilocybin and psilocin when using HPLC with chemiluminescence detection. HPLC has also been employed to detect active compounds in human blood (Hasler et al., 1997; Lindenblatt et al., 1998) and urine samples (Hasler et al., 1997).

Alternatively, various gas chromatography techniques combined with mass spectrometry have been used to detect and quantify psychedelic compounds in both mushrooms (Casale, 1985; Repke et al., 1977; Strižný et al., 2003) and humans (Sticht and Käferstein, 2000), while capillary zone electrophoresis has also been suggested for such purposes (Pedersen-Bjergaard et al., 1997). Compared to LC methods, capillary electrophoresis methods have several advantages, including low solvent utilization, smaller necessary sample volumes, high separation efficiency, and shortened analysis times (Wieczorek et al., 2015). However, Kolaczynska et al. (2021) developed a LC with tandem mass spectrometry (LC-MS/MS) method, with enhanced sensitivity and short run time, that can be adjusted to analyze human plasma or urine for various tryptamines, including psilocin, 4-hydroxyindole-3-acetic acid, or psilocin glucuronide using smaller amounts of sample. Thus, optimized methods of bioactive psychedelic mushroom metabolites are continuously being developed.

4. Detection through urine

Upon ingestion of psychedelic mushrooms, psilocybin dephosphorylates to psilocin, some of which is converted to psilocin glucuronide, a metabolite that is detectable in urine (Manevski et al., 2010; Shoda et al., 2011). To date, accepted forensic and pharmacokinetic analyses of psilocin glucuronide in urine are based on detection by TLC, HPLC, and LC-MS, which represent valid methods of detection for forensic analysis and pharmacokinetic studies (Gartz, 1989a, 1989b; Martin et al., 2014; Shoda et al., 2011). Psilocin glucuronide is more stable than psilocin and thus represents a more reliable indicator of psychedelic mushroom usage (Martin et al., 2014). For detection, psilocin can be obtained from psilocin glucuronide through hydrolysis using *Escherichia coli* derived β -glucuronidase (Kamata et al., 2006). Although psilocin glucuronide has been enzymatically produced using pretreated rat liver microsomes (Shoda et al., 2011) and chemically synthesized using precursors from microbes (Martin et al., 2014), this compound remains commercially unavailable as reference material (Martin et al., 2014; Shoda et al., 2011). The detection of other metabolites derived from psychoactive mushrooms are often used in forensics and research.

4.1. DNA-based methods offer an alternative approach

DNA-based methods offer an alternative approach to detecting the presence of psychedelic mushrooms. Rather than detecting secondary metabolites, this approach detects specific genetic markers associated with psilocybin bioproduction (Nugent and Saville, 2004; Zhang et al., 2021). Such methods can overcome setbacks related to the previous methods described, including the degradation of active metabolites,

when identifying psychedelic mushrooms. Psychoactive metabolites are prone to degradation with exposure to UV radiation (Rafati et al., 2009) and extreme temperatures (Gotvaldová et al., 2021). Identification of tryptamines in pulverized mushrooms can also be difficult (Stebelska, 2016), since rapid degradation or active compounds also occurs when products are powdered and stored (Gotvaldová et al., 2021). However, DNA-based methods cannot be used for detection or quantification of substances in plant or mammalian samples (Stebelska, 2016). Therefore, DNA-based approaches can be used for forensic analyses but contribute little value to clinical research.

4.2. Metabolism in humans, pharmacokinetics, and pharmacodynamics of bio-active components

Psilocybin and its related tryptamines have a rich diversity in biological activity which can, themselves, result in psychoactive effects, contribute to perpetuating or enhancing psychoactive effects, or are non-psychoactive. Generally, psychoactive substances are defined as those that stimulate or depress the central nervous system to elicit hallucinations, alter mood or perception, or disturb motor functions (Barratt et al., 2017). Anecdotal and preliminary research suggest that the biological outcome of mushrooms is not strictly a result of psilocybin and that there are other compounds that interact to alter the psychoactive and/or medicinal experience. Gartz (1989a, 1989b) evaluated data from 24 cases of accidental ingestion of *Inocybe aeruginascens*, a species of mushroom with high concentrations of aeruginascin and baeocystin (Gartz, 1989a, 1989b). Of particular note, Gartz (1989a, 1989b) reported that in all cases of accidental ingestion, patients only reported a euphoric response, in contrast to the frequent reports of slight, and sometimes deep, dysphoric outcomes from people who ingested mushrooms with high levels of psilocybin and psilocin without aeruginascin. The author concluded that aeruginascin can modify the pharmacological action of psilocybin. In contrast, they also reported that use of mushrooms with high levels of baeocystin, such as *P. semilanceata*, lead to higher rates of dysphoric experiences (Gartz, 1989a, 1989b). More recent studies have further supported the importance of these interactions in mushroom extracts. The implications of such studies are expanded upon in this section.

5. Different tryptamines and the entourage effect

Two major hallucinogen chemical classes exist and are categorized as mescaline-like, or psilocybin-like, respectively based on their phenethylamine or tryptamine backbones (Fantegrossi et al., 2008). To date, most research efforts related to tryptamines derived from psychedelic mushrooms have focused on psilocybin and psilocin. However, this diverse group of mushrooms are known to produce an array of other secondary metabolites that can have interactional effects that alter subjective experiences upon ingestion. Included are several compounds related to psilocin with direct psychoactivity, in addition to a number of other compounds that may interact with the tryptamines and receptors in various ways that alter their biological activity.

Although psilocybin is the most well-known tryptamine of *Psilocybe* mushrooms, there are additional tryptamines present, such as baeocystin and aeruginascin (Glatfelter et al., 2022; Sherwood et al., 2020). While psychoactive properties are primarily thought to be linked to partial agonism of serotonin (5-HT) receptor 5-HT_{2A} receptors, broad spectrum activation of different receptors and receptor subtypes could be responsible for the variable psychedelic effects reported by different psychedelic mushroom users (Geiger et al., 2018). Additionally, various compounds can collectively contribute to what is known as “the entourage effect”, which describes the synergistic interaction of a variety of different metabolites that enhance the activity of the primary active components (Russo, 2019). Tryptamine concentrations and profiles across different *Psilocybe* species are highly variable. Thus, the diversity

of *Psilocybe* species results in differential production of an assortment of tryptamines in different concentrations (Glatfelter et al., 2022), which is likely responsible for variable psychoactive effects of different mushrooms. Since various psychedelic mushrooms produce an extensive array of tryptamine derivatives, it has been suggested that different combinations of these, and thus alternative species of mushrooms could be more of interest for certain applications (Matsushima et al., 2009).

It is also relevant to indicate that psilocybin is a prodrug of dephosphorylated psilocin, which is largely responsible for the psychedelic effects (Fantegrossi et al., 2008; Helander and Beck, 2008; Passie et al., 2002). Similarly, it is postulated that baeocystin dephosphorylates to form norpsilocin whereas methylation of aeruginascin yields the bioactive 4-HO-TMT (Fig. 2) (Chue et al., 2022). It is likely that structural modifications of tryptamines into various tryptamine derivatives have variable bioactivities in humans. Although additional evidence is needed to confirm some of the following postulations, human metabolism of baeocystin, psilocybin, and aeruginascin into their respective derivatives norpsilocin, psilocin, and 4-HO-TMT could indicate that the former are prodrugs of the latter. Thus, psychotropic effects of psychedelic mushrooms are associated with metabolic activity and likely the combined action of numerous tryptamines, which demonstrates different synergizing entourage effects. The activities of various tryptamines and related compounds are described in the following subsections.

6. Psychoactive dose ranges

Psilocybin is mainly water soluble, whereas psilocin is more soluble in lipids (Wieczorek et al., 2015). When psilocin plasma levels reach 4–6 µg/mL in humans, psychological effects commence, with a threshold oral dose of 3–5 mg resulting in sympathomimetic effects, while full psychedelic effects arise with 8–25 mg oral doses of psilocybin (Passie et al., 2002). Brown et al. (2017) demonstrated a linear pharmacokinetic trend of psilocin over a two-fold range of 0.3–0.6 mg/kg. Detectable amounts of psilocin in human plasma can be found within 20 to 40 min after oral administration, with maximum plasma concentrations occurring 70 to 100 minutes following oral administration (Dinis-Oliveira, 2017; Passie et al., 2002). Psilocybin and psilocin have respective half-lives of roughly 160 and 50 min, which generally allows psychoactive effects to dissipate in the range of 4–6 h, depending on dosage (Dinis-Oliveira, 2017).

7. Pharmacokinetics and synergistic compounds

Pharmacokinetic animal models have revealed that oral administration of psilocybin results in roughly 50% absorption followed by uniform bodily distribution, though localization to the brain corresponds to psychedelic effects (Dinis-Oliveira, 2017). Upon ingestion, dephosphorylation of psilocybin to psilocin occurs, which allows it to cross the blood-brain barrier. Dephosphorylation can be facilitated by alkaline phosphatase or esterases in the liver, intestine, or kidneys (Dinis-Oliveira, 2017; Lowe et al., 2021). Psilocin then undergoes further metabolism, involving demethylation and oxidative deamination facilitated by monoamine oxidase (MAO) or aldehyde dehydrogenase (ALDH) to yield 4-hydroxy-indole-3-acetaldehyde which can be converted to 4-hydroxy-indole-3-acetic acid or to 4-hydroxytryptophol (Fig. 2), and based on certain literary works (Dinis-Oliveira, 2017; Helander and Beck, 2008), we propose that aldehyde dehydrogenase and aldehyde reductase could perhaps respectively catalyze these conversions. However, psilocin remains rather resistant to MAO deamination compared to other mushroom tryptamines and other psychedelic compounds like dimethyltryptamine (DMT) that can lack oral activity and have shorter psychoactive timeframes (Sherwood et al., 2020). This is perhaps due to the occurrence of β-carbolines, which can inhibit tryptamine degradation. Thus, the effects of psilocin can be intensified by the presence of MAO-inhibitors (Dinis-Oliveira, 2017).

β -carbolines are a class of alkaloid compounds found in *Psilocybe* spp. (Blei et al., 2020) that can inhibit MAO-mediated degradation of serotonergic psychedelics (Halberstadt et al., 2008). While the psychoactive effects of ingesting psychedelic mushrooms can last several hours (Dinis-Oliveira, 2017), the effects of dimethyltryptamine (DMT), an additional serotonergic psychedelic, has a fast onset and short duration when smoked as a purified compound (Cakic et al., 2010). Oral ingestion of purified DMT is not psychoactive due to the degradative action of gut enzymes (Bilhimer et al., 2018). However, ayahuasca is a traditional psychotropic beverage brewed from a mixture of plant leaves that produce DMT (typically *Psychotria viridis*), along with the *Banisteriopsis caapi* vine that contains β -carbolines that prevent degradation of DMT and result in a strong and long lasting effect (Hosanagar et al., 2021). Unlike the ephemeral effects of pure DMT that can subside within minutes, the effects of ayahuasca can extend numerous hours (Chi and Gold, 2020). Although the act of smoking certain substances is, in itself, associated with faster onset and shorter persistent effects compared to oral ingestion (Borodovsky et al., 2016), the presence of β -carbolines in ayahuasca play important roles related to psychoactivity (Bilhimer et al., 2018). Since the psychedelic experience of ingesting ayahuasca or *Psilocybe* mushroom can both persist for hours, it is thought that β -carbolines could also contribute to extend the psychedelic experience of *Psilocybe* mushrooms (Lenz et al., 2021).

As mentioned, the non-psychoactive β -carbolines produced by psychedelic mushrooms can inhibit MAO-catalyzed degradation of neuroactive amines, such as serotonin in a similar fashion to the MAO inhibitors regularly prescribed to treat depression (Kim et al., 1997). Recently, the β -carbolines harmine, norharmine, and harmine have been detected as naturally occurring compounds in several species of *Psilocybe* (Lenz et al., 2021), including *P. cubensis*, *P. mexicana*, *P. cyanescens*, and *P. semilanceata* (Blei et al., 2020). Due to variable selectivity of different β -carbolines, their efficacy as MAO inhibitors may vary (Glover et al., 1982; Miralles et al., 2005) and their therapeutic value has been of interest for decades. Many commonly prescribed antidepressants are MAO-A inhibitors (Nolen et al., 1993), while MAO-B inhibitors have been used as drugs for the treatment of neurological disorders (Yamada and Yasuhara, 2004). Classic in vitro enzyme studies by Glover et al. (1982) and Buckholtz and Boggan (1977) respectively showed that harmine potently inhibits both the human forms of MAO-A and MAO-B, and harmine is a strong inhibitor of brain MAO activity. Although many β -carbolines can bind both MAO-A and MAO-B, generally they more selectively bind MAO-A types (Miralles et al., 2005). Harmine and harmine are both strong and reversible inhibitors of MAO-A (Lenz et al., 2021), which is the main MAO-subtype responsible for catalyzing the inactivation of psilocin to 4-hydroxyindole-3-acetaldehyde, inhibition of which would increase or extend the psychoactive effects of psilocin (Lenz et al., 2021). Thus, both β -carbolines and tryptamines use L-tryptophan as their precursor but are derived from independent biosynthetic pathways (Fig. 2). Although both classes of molecules have different pharmacological targets, they may act synergistically to intensify and perpetuate psychotropic effects (Lenz et al., 2021).

Psilocin is also metabolically eliminated in the form of psilocin glucuronide, with up to 80% of psilocin testing as the glucuronide conjugate 5-h following intoxication (Kamata et al., 2006). This indicates a phase II pathway for psilocin metabolism. The process of glucuronidation involves the addition of a glucuronide moiety to a compound, forming highly hydrophilic, negatively charged products which are selectively transported to the bloodstream or to organs of elimination (Yang et al., 2017a). Psilocin glucuronide arises from the addition of a glucuronide moiety, catalyzed by UDP-glycosyltransferases (UGTs) (Manevski et al., 2010). Glucuronidation facilitated by UGTs represents a major metabolic conversion that occurs in humans for a variety of different drugs (Järvinen et al., 2022) and signifies defense or detoxifica-

tion processes that often result in elimination of specific substances through bile or urination (Yang et al., 2017a).

8. Pharmacodynamics of psychedelic mushroom tryptamines

Psilocin has structural similarities to 5-HT, with a comparable process of human metabolism (Dinis-Oliveira, 2017). 5-HT-like tryptamines can bind and activate both serotonergic and non-serotonergic receptors (Fantegrossi et al., 2008). However, psilocin mainly acts as an agonist to 5-HT receptors (Carter et al., 2005; Markopoulos et al., 2022) and has the highest affinity for the 5-HT_{2A} subtype (Bogadi and Kaštelan, 2021; Wittmann et al., 2007) and lower affinity for 5-HT_{1A} subtype (Passie et al., 2002; Wittmann et al., 2007). It is posited that much of the psychedelic effects of serotonin-like hallucinogens are linked to 5-HT_{2A} receptor activity (Fantegrossi et al., 2008; Geiger et al., 2018). Ketanserin is a selective 5-HT_{2A} receptor antagonist (Quednow et al., 2012) used as an antihypertensive drug. Pre-administration of ketanserin before psilocybin results in attenuated [92,93] or full inhibition of psychedelic effects (Passie et al., 2002). Pre-treatment with the dopamine D2 receptor antagonist haloperidol also reduces the psychedelic effects of psychedelic mushrooms, while the activity of psilocybin significantly limits binding of another dopamine receptor antagonist, raclopride, to the dopamine D2 receptor (Vollenweider et al., 1999). However, unlike lysergic acid diethylamide (LSD), that targets both 5-HT_{2A} (Nichols, 2016) and D2 dopamine receptors (Giacomelli et al., 1998), psilocybin and psilocin possess no affinity for dopamine D2 receptors (Passie et al., 2002). Since serotonergic and dopaminergic systems can play influential roles in modulating one another (Esposito et al., 2008), there is likely an indirect response related to psilocybin and psilocin in the modulation of dopamine D2 receptor function and/or via enhanced dopamine transmission.

When investigating the pharmacodynamics of *Psilocybe* tryptamines, influence on brain activity can involve increased cortical metabolism in various parts of the brain, and to a lesser degree in the sensorimotor and optical cortices (Passie et al., 2002). Psilocybin administered to mice elicits behavioural responses possibly reflective of hallucinogenic effects, yet the impacts of baeocystin were indistinguishable from the saline control (Sherwood et al., 2020). Although baeocystin dephosphorylation upon metabolism results in the emergence of the full 5-HT_{2A} agonist norpsilocin, it is thought that norpsilocin becomes inactive by MAO metabolism before its target is reached and/or norpsilocin may be unable to pass the blood-brain barrier. Thus, the lack of psychoactive effects of norpsilocin is expected to be the result of pharmacokinetics, rather than lack of agonist activity at the 5-HT_{2A} receptor (Sherwood et al., 2020). Further research is needed to determine the biological effects of baeocystin delivered as a whole mushroom with MAO inhibitors and other metabolites present.

Another tryptamine, aeruginascin, represents the quaternary ammonium conjugate of psilocybin (Lenz et al., 2021). Experiments using synthetic precursors in aqueous acidic conditions have confirmed that aeruginascin can be hydrolyzed to its active form, 4-HO-TMT (Fig. 2) (Chadeayne et al., 2020). 4-HO-TMT shows low binding affinity for many of the 5-HT-family receptors and is also unlikely to cross the blood-brain barrier. Specifically, 4-HO-TMT shows no activity at 5-HT₃-type, but stronger affinity for 5-HT₂-type receptors (Chadeayne et al., 2020), although more than six times less binding affinity for 5-HT_{2A} than psilocybin is expected. 5-HT₂ receptor activation is associated with decreased anxiety (Filip and Bader, 2009) and aeruginascin ingestion alone results in mood boosts without psychedelic effects, perhaps due to lower 5-HT binding affinity and/or limited occurrence in the brain compared to other more psychoactive tryptamines. The occurrence of aeruginascin has been associated with more pleasant psychedelic mushroom experiences (Gartz, 1989a, 1989b) and more research is needed

to determine the importance of aeruginascin and 4-HO-TMT as they relate to modifications in pharmacological action.

9. Health implications of psychedelic mushrooms

5-HT receptors play key roles in numerous physiological and behavioural conditions, including anxiety, mania, drug addiction, depression, schizophrenia, autism, and obesity (Filip and Bader, 2009). There is an increasing body of literature surrounding the efficacy of psilocybin-producing mushrooms for treating 5-HT-related illnesses (Bogadi and Kaštelan, 2021; Bogenschutz and Pommy, 2012; Carhart-Harris et al., 2017; Fraga et al., 2021; Glynos et al., 2022; Grob et al., 2011; Lugo-Radillo and Cortes-Lopez, 2021; Moreno et al., 2006; Wilcox, 2014; Wittmann et al., 2007). Additionally, 5-HT_{2A} receptors are linked to immune function, and treatment with psilocybin can induce various immune factors involved in immune system activation (Bove and Mokler, 2023). Although there is currently no direct evidence, the 5-HT agonism of secondary metabolites of *Psilocybe* spp. have been hypothesized as a preventative and/or treatment approach for COVID-19, since selective 5-HT reuptake inhibitors (SSRIs) can reduce SARs-CoV-2 viral load (McKernan et al., 2021a). Recently, an a priori prediction termed the psilocybin-telomere hypothesis has been proposed, stating that psilocybin consumption could have beneficial effects on genetic aging by positively impacting the telomere length of leucocytes (Germann, 2020).

More evidence must be gathered to validate the medicinal uses of psychedelic mushrooms. However, evaluating safe parameters for different treatment measures involves a comprehensive body of literature for sequential investigations (Grob et al., 2011). Ultimately, the regulated process of continuous clinical trials would allow the highest likelihood of psilocybin achieving more societal credibility as a viable medicinal compound (Rucker and Young, 2021). It has been reported that the vast majority of individuals that use illicit drugs could be considered non-problematic users, with casual users being able to sustain non-problematic, moderate consumption with no hazard to personal, social or public health (Móro et al., 2011). In the United States of America, a minimum of 7.4 million people used psychedelics in 2021 (SAMHSA, 2022). Corrigan et al. (2022) conducted a survey to determine the attitudes of mental health service users in the Tallaght area of South Dublin, Ireland toward psychedelics and psilocybin-assisted therapies. They reported that the majority (72%) support further research related to psilocybin-assisted therapies, while more than half (59%) support the use of psilocybin as a therapeutic (Corrigan et al., 2022). Thus, continuous research efforts associated with the medicinal value of psychedelic mushrooms will enhance social acceptance and emerging research shows potential to treat an extensive array of ailments, from mental illnesses to microbial infections and beyond. The following section reviews pertinent information associated with medicinal uses of psychedelic mushrooms.

10. Moral and emotional outlook

Human utility of psychedelic mushrooms also relates to autognostic utility, promoting neurocognitive, social, and psychological self-enhancement (Móro et al., 2011). Occasional use of psychedelics can facilitate mystical-type experiences that help some enhance life satisfaction, personal well-being, and positive behavioural changes (MacLean et al., 2012). In this regard, mystical experiences are multifaceted and must incorporate multiple factors such as feelings of unity, positivity, sacredness, and spatial/temporal transcendence that combine to form a single experience (Barrett et al., 2015). Psilocybin administered under supportive settings can result in greatly meaningful and spiritually significant experiences that increase personal well-being and life satisfaction for up to 14-months (Griffiths et al., 2008). This persistence of positivity, commonly referred to as the “psychedelic afterglow”, has been shown to result in higher energy levels, improved interpersonal, and

a decrease in substance use until it gradually tapers off (Johnson et al., 2014; Majić et al., 2015). However, certain fundamental realizations arising from mystical-type mushroom experiences can remain with the user indefinitely. Medium and high doses of psilocybin lead to loosening of ego-boundaries, dreaminess, feelings of unification with a “higher reality”, profound changes in perception, followed by insightfulness after the effects of psilocybin had worn off (Hasler et al., 2004). Pokorny et al. (2017) noted acute effects of psilocybin allowed enhanced emotional empathy related to changes in subjective perceptions, though they found no changes in cognitive empathy or moral decision making. While additional research should be completed, psychedelic mushrooms show potential to enhance certain moral and empathetic perceptions of the users, which could have positive social implications. The occurrence of deep psychospiritual episodes can be associated with therapeutic results (Grob et al., 2011).

11. Palliative care

An important and increasingly used application of psilocybin therapy relates to palliative care for patients with terminal illnesses. Although existential distress is common in individuals with terminal illnesses, it is often unrecognized and/or challenging to address. Such psychological turmoil can negatively impact individuals on multiple levels (e.g., personal, physical, relational, spiritual) and can intensify pain, depression, and anxiety, ultimately leading to the desire for accelerated death (Mayer et al., 2022). Although early research in the 1960s outlined the potential of psychedelics to treat existential conflicts associated with advanced-stage cancer, follow-up work has largely been set back over the 4–5 decades that followed (Grob et al., 2011; Ross et al., 2016). Grob et al. (2011) studied the effect of moderate psilocybin doses (0.20 mg/kg) on patients experiencing anxiety from advanced-stage cancers. Two treatments (one being psilocybin, the other being 250 mg of niacin as a placebo) were administered several weeks apart. The order of treatment delivery was randomized and unknown to the participants. They found no significant changes in state anxiety 2-weeks following treatment, but substantial, though non-significant, reduction in state anxiety in the 6-h that followed treatment. Alternatively, a significant reduction in trait anxiety was documented 1-month after the second dose that persisted for 6-months. Moreover, based on the Beck Depression Inventory (BDI) and Profile of Mood States, mood improved for 2-weeks following treatments, but only reached significance 6-months after based on BDI. Although this study showed promising trends, the authors identified the need for additional studies to validate their findings due to variability in the follow-up protocols and the moderate psilocybin doses used (Grob et al., 2011).

Ross et al. (2016) conducted a similar study using a slightly higher dose of psilocybin (0.30 mg/kg) in conjunction with psychotherapy on cancer patients diagnosed with anxiety and/or depression. Similar to the Grob et al. (2011) study, doses of either psilocybin (0.30 mg/kg) or niacin (250 mg control) were randomly ordered and given an average of 52-days apart. The authors reported that both trait and state anxiety were significantly reduced from 1-day to 7-weeks after the first dose and BDI also showed significant differences between experimental and control groups in the 1-day to 7-weeks following the initial dose. After the second dose, there were no significant differences in BDI between treatment groups throughout the 6.5-months that followed. Similarly, there were no significant differences between populations related to state and trait anxiety at 2.5- and 6.5-months, respectively. However, state anxiety of the experimental population showed significant reductions again at 6.5-months after the second dose. The authors reported the experimental population to occasion mystical-type experiences from the moderate doses used that significantly facilitated the reduction in anxiety and depression symptoms over the timespan before the second dose (Ross et al., 2016). Taken together, these studies show that there is greater potential to treat anxiety and depression related to exis-

tential distress when moderate doses of psilocybin are combined with psychotherapy and that the time frame between doses might also impact efficacy. The setting in which drugs are taken can substantially impact the subjective experience of the intoxicated individual (Hart, 2021). Thus, having a psychedelic experience in a positive setting, with the support of a psychotherapist can dramatically influence potential for positive outcomes. Although more research associated with psilocybin-assisted therapies for palliative care is emerging, there are many questions that remain unanswered. A method for training care providers must be developed, requirements for accessing psilocybin should be determined, and associated costs must be evaluated (Mayer et al., 2022). Much more research is needed to determine the efficacy of different doses and to optimize timeframes and quantities of sequential dose delivery. Currently, the National Cancer Institute is conducting four studies involving psilocybin in relation to cancer treatment (National Cancer Institute, 2022).

12. Therapeutic dosages and expert support

Although higher doses of psilocybin are generally associated with mystical-type experiences (MacLean et al., 2011), and such experiences are associated with positive outcomes (Barrett et al., 2015; Ross et al., 2016), the occurrence of complete mystical experiences are not always required to elicit positive effects. Nicholas et al. (2018) conducted a study to examine the impact of successive increasing dosing of psilocybin on the subjective positivity of healthy patients. The experiment included moderate and high dosages that were delivered under the supervision of trained guides, an average of 4-weeks apart at increasing quantities (0.30 mg/kg, 0.45 mg/kg, and 0.60 mg/kg). The guides also provided preparatory counseling to the participants prior to the first psilocybin dose. There were no significant dose-dependent differences in the occurrence of complete mystical experiences, although the authors did describe significant dose-dependent differences in space and time transcendence. Ultimately, it was reported that complete mystical experiences may not be essential for positive outcomes and that the impact of interval on positive outcomes of consecutive doses remains unknown (Nicholas et al., 2018).

Griffiths et al. (2017) studied the impact of psilocybin with meditation training on the prosocial attitudes/behaviours in addition to psychological functioning of healthy volunteers. The three experimental treatments included a very low dose (0.01 mg/kg) administered two-times with moderate spiritual support, higher doses (0.29 mg/kg and 0.43 mg/kg) separately administered both with moderate spiritual support, and again, the same higher doses (0.29 mg/kg and 0.43 g/kg) but both with high spiritual support. Sequential psilocybin/support sessions occurred roughly 1-month apart. Higher dose treatments resulted in significant acute and continuous positive effects related to gratitude, meaningfulness, interpersonal closeness, forgiveness, and spiritual transcendence. No differences in subjective effects were found between first and second sessions for any treatment, though subjective experiences of both high dose groups were significantly higher than the low dose group, with negligible differences between the high dose groups. Occurrence of complete mystical experiences were also higher in both higher dose populations compared to the low dose population at either session. After 6-months, both higher dose groups showed positive changes in mood, behaviour, altruism, and increased spirituality that was significantly different than the low dose group. Moreover, the higher dose group with high spiritual support experienced significantly higher altruistic and positive social behaviour and enhanced spirituality compared to the high dose group with moderate spiritual support (Griffiths et al., 2017). Ultimately, it seems that concurrent use of psilocybin and psychotherapy or spiritual support can enhance the positive implications of psilocybin-assisted therapies to a certain degree. Unfortunately, many of these studies do not explicitly list the source, or purity of the psilocybin used, variability of which could drastically

impact the results obtained due to differences in potency or entourage effects. To improve replicability and standardized dosages it is of principal importance to establish accredited repositories for psychedelic substances that can provide pure, standardized, uniform psychedelic mushroom compounds produced in accordance with GMP measures.

13. Depression and anxiety

Mental health disorders such as various forms of depression and anxiety are widely treated with prescription SSRIs (Esposito et al., 2008; Jensen, 2004; Matsushima et al., 2009; O'Donnell et al., 2019; Rocha et al., 2019). Use of psychedelic mushrooms to treat such disorders is currently an emerging research topic (Fraga et al., 2021) due to the 5-HT action of tryptamines and positive initial results. Carhart-Harris et al. (2017) tracked cerebral blood flow-related changes in brain function of patients with treatment-resistant depression. Two doses of psilocybin (roughly 0.08–0.16 mg/kg and 0.20–0.40 mg/kg) were given 1-week apart. Patients showed acute decreases of default-mode network integrity, which then post-acutely increased, followed by mood improvements. Reduction in temporal cortex cerebral blood flow bilaterality of the left amygdala and left medial temporal lobe were associated with improved outcomes. These results were potentially linked to a “reset” mechanism by which acute modular disintegration during the psychedelic state facilitates modular re-integration after the psychedelic experience that renews normal functioning (Carhart-Harris et al., 2017). Such outcomes could potentially be linked to the occurrence of the “psychedelic afterglow” previously described. A follow-up study by the group compared psilocybin (roughly 0.20–0.40 mg/kg) delivered as two separate doses 3-weeks apart to the antidepressant and anxiolytic escitalopram. All patients received psychological support. The authors reported no significant differences in effects between experimental and control populations after 6-weeks (Carhart-Harris et al., 2021). There is a potential dual mode of antidepressant action associated with psychedelic mushrooms, that is, via action at the 5-HT_{2A} receptors and MAO inhibition by β -carbolines. Ultimately, trends in psilocybin research for treating depression show very promising results, though more work in this area is needed to better understand the psychedelic and antidepressant actions of psilocybin, how these biological manifestations are linked, and to further validate efficacy (Fraga et al., 2021).

14. Obsessive compulsive disorder

The value of psilocybin in treating anxiety-related disorders such as obsessive-compulsive disorder (OCD) has also been demonstrated. Prescription SSRIs represent the standard treatment option for the majority of individuals suffering from OCD (Lugo-Radillo and Cortes-Lopez, 2021). Unfortunately, the effectiveness of this strategy depends on the patient in question and these drugs do not always represent an appropriate strategy for OCD treatment. Individual anecdotes suggest that adult males suffering from OCD who regularly self-medicate with 2-g of *Psilocybe* mushrooms over extended periods had reductions of intrusive obsessive thoughts and anxieties, resulting in improved qualities of life (Lugo-Radillo and Cortes-Lopez, 2021; Wilcox, 2014). Additionally, a formerly institutionalized adolescent male experiencing symptoms of social distance, intense anxiety, and lack of motivation consumed 2-g of *Psilocybe* mushrooms three-times over a period of 18-months, after which he more regularly joined in group activities, communicated more effectively, felt less anxious, and expressed emotion more readily (Bogadi and Kaštelan, 2021). Although anecdotal, these cases document occasions that psychedelic mushrooms showed higher efficacy than conventional treatments for alleviating symptoms of anxiety-related disorders, identifying the importance of researching alternative treatment strategies for 5-HT-related disorders.

Only a limited body of literature represents clinical and pre-clinical work attempting to verify the effectiveness of psychedelic mushrooms for the treatment of OCD. Although psilocybin can exacerbate anxiety in some, dosages ranging from very low to high can facilitate variable levels of success when treating OCD symptoms (Moreno et al., 2006). These findings were mirrored in a pre-clinical study using mice (Matsushima et al., 2009). Marble-burying behaviour of mice, which is used as a model system to study aspects of OCD was reduced by low doses of psychedelic mushrooms, while higher doses significantly limited marble-burying behaviour. These authors also mention that entourage effects of different psychedelic mushrooms could directly impact treatment efficacy (Matsushima et al., 2009). Variable tryptamine cocktails produced by different types of mushrooms could deliver inconsistent medicinal effects when compared to purified psilocybin, which could prove to be problematic when interpreting results and/or replicating experiments (Gotvaldová et al., 2022). While these studies show promising results, they only scratch the surface in presenting the evidence required to normalize psychedelic mushroom-assisted treatments for anxiety disorders like OCD. More studies must be completed to elucidate specific modes of action for refined treatment strategies that integrate multiple secondary metabolites from psychedelic mushrooms.

15. Substance use disorders

There is promising preliminary evidence for the use of psychedelic mushrooms in treating substance use disorders such as alcoholism and tobacco addiction. 5-HT system modification mediated by psilocybin treatments can enhance emotional processing and prevent the persistence of negative behaviours, perhaps by indirectly regulating the default mode network, executive control, and salience networks, which play key roles related to substance use disorders (de Veen et al., 2017). Preliminary data concerning the use of psilocybin for treating alcohol dependence have been published by Bogenschutz et al. (2015) where they examined the impact of two psilocybin doses (0.3 mg/kg and 0.4 mg/kg) administered 4-weeks apart along with motivational enhancement therapy. Results showed increases in abstinence of alcohol use following the psilocybin treatments, which justifies the demand for subsequent studies with larger sample sizes (Bogenschutz et al., 2015).

Two papers from Johns Hopkins Department of Psychiatry and Behavioural Sciences investigated psilocybin as a treatment for tobacco addiction (Garcia-Romeu et al., 2014; Johnson et al., 2014). A 15-week treatment session included 0.30 mg/kg and 0.43 mg/kg doses of psilocybin delivered at weeks 5 and 7, respectively, with the option for an additional dose (0.43 mg/kg) at week 13. A “target-quit date” was scheduled for the program’s 5th week to match up with the time of initial psilocybin delivery. 80% of patients abstained from smoking in the 6-months following the program (Garcia-Romeu et al., 2014; Johnson et al., 2014), which exceeds general success rates of other smoking cessation therapies with success rates typically ranging from 24.9–35.2% (Johnson et al., 2014). Success rates were significantly correlated with the occurrence of mystical-type experiences which occurred in 60% of the described population that abstained from smoking for 6-months (Garcia-Romeu et al., 2014). Preliminary evidence supports the notion that psilocybin may be able to enhance cognitive flexibility in animals (Torrado Pacheco et al., 2023), which could help individuals adjust their behaviours in response to change (Dajani and Uddin, 2015).

The impact of mushroom β -carbolines could play an additional role in smoking cessation. The activities of MAO-A and —B are significantly lower in tobacco smokers than non-smokers due to the inhibition of 5-HT deamination, a dose dependent result of tobacco usage (Castagnoli and Murugesan, 2004). Therefore, the act of smoking cigarettes could contribute to habit-forming actions and depression (Herraiz and Chaparro, 2006) due to increased 5-HT uptake (Castagnoli and Murugesan, 2004). Although psilocybin mediated 5-HT system modifi-

cation and mystical-type experiences have been suggested as modes of action for treating substance use disorders (de Veen et al., 2017; Garcia-Romeu et al., 2014), the studies mentioned only represent the groundwork for the research that must be carried out. Ultimately, the specific mechanisms of psychedelic mushrooms for treating substance use disorders must be elucidated to develop effective treatment options for a variety of substance dependence issues.

16. Implications of microdosing

Although the majority of pre-clinical and clinical research related to the effects of psilocybin involve moderate-to-high doses, emergent studies also relate to the mental health impacts of microdosing. Microdosing has generated high interest over the past few years and is defined as continuous, regular consumption of psychedelics in doses that are too low to alter normal human function (Anderson et al., 2019; Cavanna et al., 2022; Kaertner et al., 2021; Kuypers, 2020; Rootman et al., 2022). Anecdotal reports of microdosing psychedelics describe improvements in mood and cognition (Anderson et al., 2019; Higgins et al., 2021; Kaertner et al., 2021). It is thought that regular consumption of psilocybin microdoses over prolonged periods can positively impact mood, depression, and anxiety, and overall emotional wellbeing in a similar way as taking moderate-to-large doses with personal support. Microdosing with psychedelics can reportedly improve state anxiety, depression, emotional stability, and psychological wellbeing, however, these outcomes may be due to positive expectancy which suggests a placebo effect (Kaertner et al., 2021). Although the positive impacts of microdosing with psilocybin might be due to the expectation of beneficial effects, low doses of *Psilocybe* mushrooms can alter electroencephalogram rhythms and impart noticeable subjective effects (Anderson et al., 2019).

Since research related to microdosing with psilocybin is lacking (Cavanna et al., 2022; Rootman et al., 2022), mushroom microdosers have come up with different microdosing strategies. While some microdosers take low doses of *Psilocybe* alone, others combine low doses of *Psilocybe* mushrooms with *Herichium erinaceus* (Lion’s Mane) and/or niacin (Rootman et al., 2022). There is potential added benefit in taking low doses of *Psilocybe* with Lion’s Mane. Hericene A is a derivative of the isoindole N-de(phenylethyl)isohericerin, both of which are produced in Lion’s Mane and found to be potent promoters of axon outgrowth and neurite branching in vitro (Martínez-Mármol et al., 2023). In pig brains, a single low dose of psilocybin (0.08 mg/kg) can increase persistent synaptogenesis (Raval et al., 2021). Thus, combining the two could facilitate extension of axons, branching of dendrites, and new synaptic connections. However, Rootman et al. (2022) showed that microdosing has small-to-mid level mood and mental health benefits across different demographics, though combining psilocybin, Lion’s Mane, and niacin had no additional benefit related to mood or mental health. However, the combination of psilocybin, Lion’s Mane, and niacin was associated with improved psychomotor reactions in the older demographic tested when compared to participants taking psilocybin alone or with Lion’s Mane (Rootman et al., 2022). Niacin can potentially prevent or improve neurodegenerative diseases through various neuroprotective roles and can mediate neuronal development (Gasperi et al., 2019), which might explain some results presented by Rootman et al. (2022). Due to the axon outgrowth and neurite branching effects of Lion’s Mane and synaptogenesis impact of psilocybin, future studies involving pairing of the two should look at alternating the timeframe of each. Taking Lion’s Mane supplements alone for an extended timeframe, then microdosing with psychedelic mushrooms might enhance axon outgrowth and neurite branching for a prolonged period, then allow focused synaptogenesis between the extended cells.

17. Parkinson's disease

Abnormal MAO-A activity can be associated with psychiatric disorders like depression, whereas MAO-B irregularity can be linked to neurological disorders such as Parkinson's disease (PD). Reduced PD occurrence in tobacco users is well documented, with an estimated 30–40% lower incidence rate (Castagnoli and Murugesan, 2004; Herraiz and Chaparro, 2006). It has been suggested that the action of MAO-B in the brain can influence the production of neurotoxins that can potentially induce PD (Castagnoli and Murugesan, 2004; Herraiz and Chaparro, 2006). Since MAO-B inhibitors are used in the prevention and treatment of PD [121], and norharmane, a β -carboline produced in some psychedelic mushrooms (Lenz et al., 2021), is a competitive and reversible inhibitor of MAO-B (Herraiz and Chaparro, 2006), it is possible that mushroom β -carbolines can aid in treating PD. The implications surrounding the use of mushroom-derived β -carbolines on MAO inhibition represent a potential area of study that has not yet been pursued by pre-clinical or clinical psychedelic mushroom researchers. Hence, it is important to conduct trials relating to medicinal impacts of lesser-known psychedelic mushroom secondary metabolites.

18. Immunological implications

Bove and Mokler (2023) conducted a recent study to evaluate the impact of a single psilocybin dose (20 mg/kg) in activating various immune factor responses. Blood from female rats given a high dose of psilocybin was analyzed for different cytokines, chemokines, and peptides occurrences after 1-day and 7-days. Although the experiments showed high variability, many pro-inflammatory and anti-inflammatory cytokines in addition to other inflammatory mediators significantly increased after 1-day and continued to increase over the 7-day period. The authors also documented insignificant, though progressively increasing levels of most chemokines and vascular endothelial growth factors. The responses shown were relatable to generalized immune cell activation, and therefore not considered a specific coordinated effect due to the apparent opposite functions of the triggered cytokines (pro-inflammatory and anti-inflammatory). However, longer-term increases in the activity of certain chemokines that can support brain neurogenesis should warrant further research (Bove and Mokler, 2023). Since the authors administered extremely high doses of psilocybin to female rats via injection, it remains to be seen if similar responses can be achieved with variable dosages orally administered in animal studies, as well as in both men and women. If activation of specific immune responses can be targeted with treatments of select mushroom metabolites, it could be possible to devise treatments for any number of physiological ailments. Ultimately, this publication opens the door to a previously unexplored realm in the use of psychedelic substances.

19. Fibromyalgia

5-HT receptor subtypes play critical roles related to facilitation or inhibition of neuropathic pain at both central and peripheral levels (Liu et al., 2020). Fibromyalgia (FM) represents a chronic pain syndrome, the pathophysiology of which remains unclear (Cheng et al., 2018). FM is difficult to define, troublesome to treat, and impacts up to 4% of people globally (Glynos et al., 2022; Häuser and Fitzcharles, 2018). Though FM has been re-categorized from being a form of neuropathic pain (Cheng et al., 2018), to being a somatic syndrome (Castellanos et al., 2020), there is evidence supporting the occurrence of small fiber neuropathy in patients experiencing FM (Cheng et al., 2018). The shift from acute to chronic pain has been shown to involve changes in the structures and functions of the nervous system, including alterations in neuroplasticity, and there is literature supporting the association of 5-HT_{2A} gene polymorphisms with FM (Castellanos et al., 2020). Due to its

5-HT_{2A} agonism, ketanserin has been used to examine the role of a 5-HT_{2A} on neuropathic pain (Liu et al., 2020) in addition to psilocybin action (Johnson et al., 2018; Quednow et al., 2012). With this in mind, psilocybin and psilocin could show potential to treat chronic pain syndromes such as FM, and serotonergic psychedelics have even been proposed as a potential treatment option for chronic pain symptoms due to various mechanisms that can potentially reset functional connectivity in the brain (Castellanos et al., 2020). Glynos et al. (2022) conducted a survey to investigate the use and potential efficacy of psychedelics in the treatment of FM. In general, participants who had used psychedelics were more open to the idea of alternative treatments options, and participants who had specifically used psychedelics for personal wellbeing reported improvements over those who had used psychedelics for other reasons. However, FM scores, pain severity, and pain interference based on psychedelic use did not differ significantly (Glynos et al., 2022). Although this investigation does not hold much value in presenting the use of psychedelics for treatment of FM specifically, it does show that psychedelics can help individuals overcome personal biases to be more open to alternative options. It also portrays the medicinal significance of intent when entering psychedelic experiences. Characteristics that are specific to the individual undergoing a drug-induced experience, including mood, physiology, and preconceived ideas of the substance and expected outcomes can be of critical influence to the enjoyability of the experience (Hart, 2021). Having a positive preconception of the substance of interest and an expectation of desired outcomes can enhance the efficacy of the treatment. This indicates the importance of educating the general population about the potential therapeutic effects of psychedelics, and concurrent use of psychedelics with some form of psychosocial or spiritual guidance when using these substances as treatment options, both of which can facilitate more positive outcomes. Still, due to the potential role of 5-HT receptors linked to chronic pain, more research is needed to either support or refute the use of serotonergic psychedelics as a treatment option for such conditions.

20. Potential hazards of psychedelic mushroom usage

Although psilocybin use is generally well received by most individuals (Castro Santos and Gama Marques, 2021; Chi and Gold, 2020; Dinis-Oliveira, 2017; Grob et al., 2011; Hasler et al., 1997; Hasler et al., 2004; Wilcox, 2014), the diversity of psychedelic mushrooms available is vast and it is possible that certain species of mushrooms could produce toxins or allergens that might adversely affect specific individuals. Muscarine is a potentially harmful alkaloid produced in the fruiting bodies of *Inocybe* spp. (Kosentka et al., 2013) that targets the acetylcholine receptors, causing profuse sweating, bradycardia, and lacrimation (Brown et al., 1962). Early research found that muscarine occurrence is widely distributed among different species of *Inocybe*, which is unpredictable based on taxonomic ranking (Brown et al., 1962). More recent work suggests that loss of muscarine precedes evolutionary gain of psilocybin in psilocybin-producing *Inocybe* species and that both compounds are not found together in *Inocybe* spp. (Kosentka et al., 2013).

The evidence supporting the physical dependence and withdrawal potential of psilocybin is extremely low, though there is high tolerance (diminishing responses after repeated, regular usage) associated with short intervals between doses and cross-tolerance with LSD (Johnson et al., 2018), which might be linked with habitual use of increasing doses. Of course, this abuse potential would depend on the individual in question, since psychedelic mushrooms are generally considered non-addictive substances (Wurst et al., 2002), and have been associated with profound improvements in addictive behaviour (Garcia-Romeu et al., 2014). However, there are certain potential hazards related to taking psychedelic mushrooms. Commonly occurring adverse physical problems can include headache, nausea, fatigue, or dizziness, while transient psychological complications sometimes include anxiety, paranoia, impaired sleep, dysphoria, grief, and preoccupation with death

(Sarparast et al., 2022). Additionally, interactions with other drugs (Wurst et al., 2002), potential to induce psychosis (Hasler et al., 1997), and possible cardiovascular irregularities (Hasler et al., 2004) are factors that must be considered when using psychedelic mushrooms. This section reviews certain areas of consideration related to potential hazards of taking psychedelic mushrooms.

21. Cautionary interactional effects of psilocybin

Although minimal evidence is available, there remains potentially adverse drug–drug, drug–food, or drug–disease interactions that can arise with use of psychedelic mushrooms (Geiger et al., 2018). Risks associated with the use of psychedelics in general are substantially amplified when mixed with additional substances in unsupervised settings (Schlag et al., 2022). In spite of the fact that there is minimal association with psilocybin in conferring 5-HT toxicity due to limited intrasynaptic 5-HT release, there is potential for psychedelic mushrooms to interact with prescription drugs such as SSRIs, antidepressants, and MAO inhibitors (Schlag et al., 2022). Such interactions could additively agonize 5-HT receptors and/or inhibition MAO-A activity to ultimately play a role related to 5-HT toxicity (Yu, 2008). When co-administered with psilocybin, it has been noted that cannabis can result in intense anxiety, while amphetamines can evoke perpetuation of thought loops by which the user fails to escape specific thoughts or ideas (Geiger et al., 2018), lending themselves to negative experiences that can persist and cause adverse afterglow effects. However, the most dangerous psilocybin associated drug-drug interaction reportedly occurs with tramadol, an opioid pain medication, the combination of which is associated with increased seizure risk (Geiger et al., 2018). Still, psilocybin seems to have lower risk potential of certain adverse events when compared to other psychedelics (Sarparast et al., 2022).

22. Potential for psychosis

Psilocybin induced brain activity can impact emotional face processing (Bernasconi et al., 2014), time perception (Wittmann et al., 2007), and potentially induce psychosis and/or the onset of schizophrenia (Markopoulos et al., 2022). While serotonergic psychedelics may be relevant for the treatment of depression or addiction, they could have negative impacts on psychotic disorders like schizophrenia (Markopoulos et al., 2022; Vollenweider and Preller, 2020). The 5-HT_{2A} system has critical involvement in human emotional, social, and self-processing (Vollenweider and Preller, 2020). There is evidence connecting 5-HT_{1A/2A} systems to formation of psychotic symptoms and cognitive/behavioural deficits that are common with schizophrenia and can also cause deficits related to attention, behavioural control, time perception (Wittmann et al., 2007), and spatial working memory (Vollenweider et al., 1998). The use of psychedelics has been linked to the development of psychosis in those that are genetically predisposed to such conditions (Vollenweider et al., 1998). Specifically, psilocybin can produce psychosis-like syndromes similar to those that resemble the initial episodes of schizophrenia (Vollenweider et al., 1998; Wittmann et al., 2007). Although there are certain considerations that should be acknowledged relating to genetic pre-disposition to psychotic disorders before using psychedelic mushrooms, psilocybin-induced psychosis might offer an alternative model to examine different antipsychotics (Vollenweider et al., 1998).

23. Cardiovascular implications

Psychedelic mushrooms have been associated with temporary increases in blood pressure and heart rate (Nkadameng et al., 2020). Moreover, 5-HT₂ activity linked to serotonergic psychedelic action has been connected to heart valve disease (Hutcheson et al., 2011). 5-HT_{2A} activity is involved in up-regulation of transforming growth factor- β 1

(Xu et al., 2002), which is thought to mediate cellular changes leading to heart valve disease (Hutcheson et al., 2011). Although 5-HT_{2A} activity may contribute, in part, to certain types of heart valve disease, additional secondary signaling pathways beyond those of 5-HT_{2A} involvement are necessary (Xu et al., 2002). It has thus been suggested that long-term, regular exposure to 5-HT₂-activating substances is needed to induce such adverse effects (Rudin et al., 2022). Alternatively, an in vitro study showed that psychedelic mushroom extracts could have potential cardioprotective properties (Nkadameng et al., 2020). Nevertheless, it is suggested that individuals with cardiovascular conditions exercise caution when using psychedelic mushrooms (Hasler et al., 2004).

24. Current strategies and future perspectives related to production of *Psilocybe* spp. and metabolites of interest

Conventional cultivation of *Psilocybe* spp. has long been practiced by humans around the world and remains a popular pastime for many individuals. In addition to conventional production practices, secondary *Psilocybe* metabolites have also been produced using chemical reagents, synthetic biology and heterologous systems to obtain isolated, purified compounds for research and industrial applications. Although synthetic psilocybin produced to GMP standards currently represents the benchmark for clinical research, using psilocybin in isolation would not produce any entourage effects (beneficial, or detrimental), and cannot accurately depict the effects of ingesting whole mushrooms. Therefore, it is important to outline how *Psilocybe* mushrooms are currently produced, and outline knowledge gaps related to production strategies to help guide future optimized production and processing practices for whole mushrooms that comply with GMP standards. In this section, the former mentioned methods are described in addition to perspectives related to CRISPR-based methods and Machine Learning (ML)-assisted strategies as potential tools for metabolic engineering of psychedelic mushrooms.

25. Conventional cultivation of *Psilocybe* spp.

Although in many places prohibition of psychedelic mushrooms has been enforced since the 1970s (Fricke et al., 2019), enthusiasts and amateur mycologists have continued to collect naturally occurring *Psilocybe* spp. and cultivate them for personal use or sale on the illegal drug market (Redhead et al., 2007). The household cultivation of psychedelic mushrooms has exploded from a niche trend into a global underground pastime (Fricke et al., 2019), creating an international community of those passionate about *Psilocybe* mushrooms. Of the cultivated *Psilocybe* spp., various isolated genetic lineages of *P. cubensis* seem to be of high interest for production (Van Court et al., 2022) and most widespread due to ease of cultivation and high psychoactive potency. However, additional cultivated species include *P. semilanceata*, *P. tampanensis*, *P. azurescens*, *P. ovoideocystidiata*, *P. natalensis*, and *P. allenii*, in addition to various *Panaeolus* species. Although there have been numerous biotechnological advancements in psilocybin production using enzymatic synthesis (Fricke et al., 2017), commercial reagents (Hofmann et al., 1959; Kargbo et al., 2020; Nichols and Frescas, 1999; Shiota et al., 2003), and heterologous systems (Adams et al., 2019; Fricke et al., 2020), conventional production of *Psilocybe* fruiting bodies remains relevant. While there might be slight differences in optimal production practices and associated factors between different *Psilocybe* species due to variable growth habits, in general, growing *Psilocybe* mushrooms is fundamentally similar to typical production practices of other Agaricales. However, the relative amounts of psychoactive metabolites derived from *Psilocybe* mushrooms can vary with growth conditions such as the presence of glucose and ammonium succinate, pH, level of oxidation during growth (Casale, 1985), substrate nitrogen, phosphate, and amino acids concentrations, climatic conditions, and developmental stage. Additionally, the developmental stage of the mushroom at the

time of harvest can impact the final potency (Gotvaldová et al., 2022). Thus, biological samples collected at different life stages or from spatially separated sections of the colony can elicit variable chemical profiles (Bigwood and Beug, 1982; Casale, 1985; Rafati et al., 2009). Demmler et al. (2020) determined that most of the genes related to psilocybin production are downregulated in mature fruiting bodies compared to carpophore primordia, which showed 395-fold higher expression of the gene encoding PsiD, which converts L-tryptophan to tryptamine (Demmler et al., 2020). This perhaps contributes to the higher potency of fruiting bodies harvested prior to breaking of the veil, as reported by *Psilocybe* mushroom growers, but further research including chemical quantification is needed to verify this assumption.

There may only be slight differences relating to optimal inoculation techniques, substrates, incubation time, and atmospheric components among different *Agaricales* (Sánchez, 2010). Below, we describe conventional methods of producing *P. cubensis*, including a schematic representation (Fig. 3). When it comes to conventional production of *Psilocybe* spp., initiating a culture begins with spore germination (Fig. 3, A). This important step leads to the development of fungal hyphae from spores. This process is physiologically characterized by the spore swelling, followed by the protrusion of the germ tube (Gottlieb, 1950). Though an isolated mycelium culture can be derived from carpophore tissue samples, or other mycelial cultures (Sánchez, 2010), which we respectively deem cloning and proliferation, spore germination differs

in its ability to use stored reserves to initiate metabolic processes (Gottlieb, 1950). Germination can be accomplished on common media such as potato dextrose agar with dextrose concentrations up to 2% (w/v) (Memon et al., 2017), suspension broths (Curtis, 1964), or damp solid media (Kumla et al., 2020). However, with water being the primary requirement and only essential component for spore germination (Gottlieb, 1950), if spores are obtained from an unaccredited source, we recommend sterilized distilled water to promote germ tube emergence to mitigate microbial contaminants (Fig. 4, A), since decontaminated spores are largely unavailable. Once hyphae emerge, there are three major media alternatives that can be inoculated to allow mycelium development (Fig. 3, B). These options include semi-solid media (culture plates solidified with a gelling agent) (Fig. 4, B), suspension broths (liquid media) (Fig. 4, C), and grain spawn cultures (solid media) (Fig. 4, D). Once colonized, sub-culturing of mycelium for proliferation can be executed on fresh media of the same type, or between those media types mentioned. The use of potato-dextrose agar is a well-documented semi-solid media option for mycelial development (Dhanasekaran et al., 2020; Montiel et al., 2008; Yang et al., 2017b), while malt extract peptone (Blei et al., 2018; Lenz et al., 2017), and Corn Meal Agar (Montiel et al., 2008) are less commonly used alternatives. Potato dextrose broth (Dhanasekaran et al., 2020; Yang et al., 2017b) and malt broths (Helbling et al., 1993) facilitate mycelial growth in liquid cultures. Mycelium derived from either semi-solid or

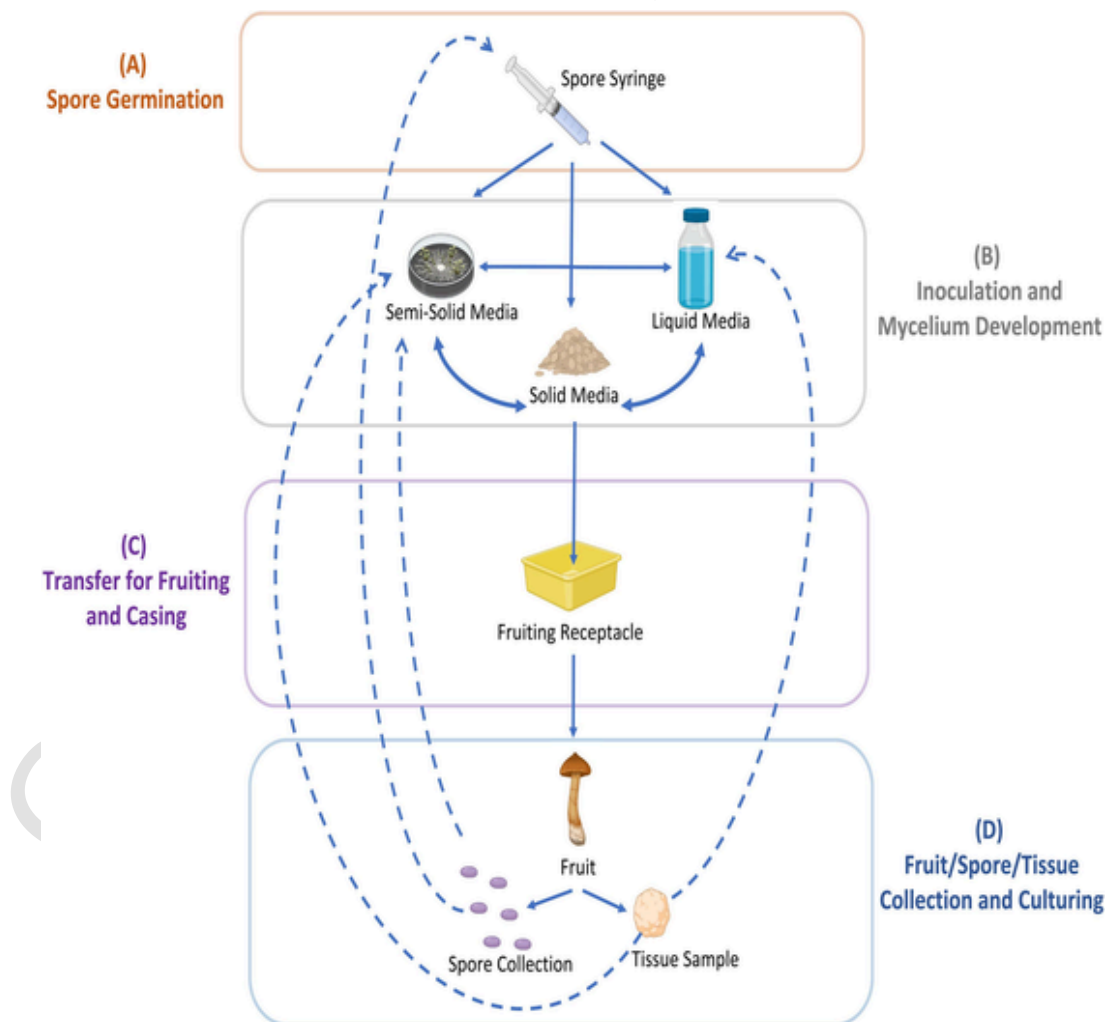


Fig. 3. Schematic representation of conventional cultivation process for *P. cubensis*. Common steps include (A) spore germination, (B) inoculation on specific substrate for mycelium development, (C) transfer to fruiting receptacle and casing, and (D) collection of fruiting bodies, spores, and fresh tissue for further culturing. Parts of the scheme were generated using [BioRender.com](https://www.biorender.com).

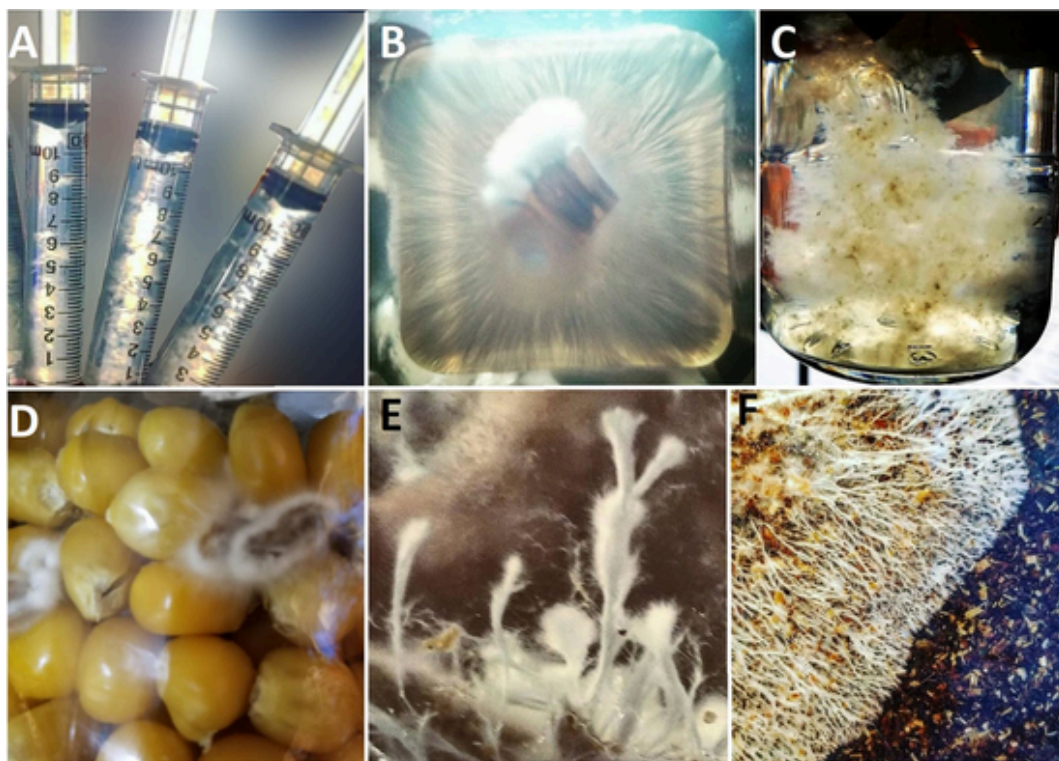


Fig. 4. Examples of *Psilocybe cubensis* mycelium at various life stages throughout the conventional production process. (A) germinated spores prepared in liquid culture syringes for inoculation. (B) Semi-solid media colonized by *Psilocybe cubensis* mycelium. (C) *Psilocybe cubensis* mycelium developing in liquid culture. (D) Mycelium colonizing solid media to make grain spawn culture. (E) Rhizomorphs stretching into open air. (F) Mycelium colonizing casing substrate.

liquid cultures can be used to inoculate grain to make spawn cultures in preparation for fruiting.

Rye, wheat, and millet are widely used as grain spawn substrates (Sánchez, 2004; Smith et al., 2002), which are usually soaked, simmered, or pressure-cooked to imbibe water, before exterior drying. Grains can be used alone, mixed with each other, or have added wheat bran, gypsum, coco coir, or sawdust to adjust nutrient content, pH, or moisture composition. These types of media allow water and nutrients to remain bound within the grains, while maintaining continuous gas phases between the grains to facilitate gas exchange (Berovic, 2019). Thus, mycelium can respire continuously while accessing stored carbohydrates and essential elements by colonizing grains that create a solid particulate matrix with expansive surface area. The specific nutrient air blend afforded by grain spawn media can facilitate rhizomorphic mycelial growth (Fig. 4, E). Rhizomorphs are thick, thread-like mycelial formations that grow from colonized substrate into aerated areas (Takahashi and Degawa, 2011). Mature rhizomorphs are hollow structures (Redfern, 1973) that develop from mycelial air pores, implicating their strategic role relating to gas exchange (Pareek et al., 2006), perhaps acting as oxygen conduits to enhance aerobic activity. Under optimal conditions, these specialized structures can grow at an accelerated rate compared to hyphal tips (Pareek et al., 2006; Rishbeth, 1968) and are associated with specimen vigor when observed in *P. cubensis* cultures.

Fully colonized grain spawn cultures can then be used for fruiting. Typically, this involves transfer of grain spawn to a receptacle (Fig. 3, C) allowing increased surface exposure to air for enhanced gas exchange and a slight drop in humidity that helps to facilitate fruiting. However, increased air exposure potentiates the likelihood of contamination and dehydration. Although too much water can hinder aerobic activity, adequate moisture content must be maintained to allow hydration and steady flow of nutrients for basidiocarp development (Bellettini et al., 2019). A common technique to inhibit contamination

while maintaining surface moisture content at an appreciable level is referred to as casing. This technique involves layering a pH-neutral, moisture-retentive substrate over the grain spawn and maintaining the spawn in a dark, unventilated, room-temperature location of high humidity for roughly one week as the mycelium colonizes the casing layer (Fig. 4, F) (Antonio, 1971). Common casing substrates include coco coir, vermiculite, or soil mixtures containing peat, clay, or loam. Basidiocarp induction occurs with development of hyphal knots, which progress into fruiting body primordia (pins) (Fig. 5, A), triggered by increasing the speed of air circulation around the colonized casing layer, which allows lower CO₂ levels, reduced temperature, and relative humidity (Baars et al., 2020). It is beneficial to keep humidity at roughly 85%, while maintaining fresh air exchange to provide O₂ for the developing fruiting bodies (Bellettini et al., 2019). By precisely manipulating humidity and air exchange using specific casing substrates, ventilation techniques, and watering regimes, growers can control pinning rate and fruiting body development (Baars et al., 2020). Light is also considered an important factor for basidiocarp development of *Agaricales* (Aghajani et al., 2018), playing roles related to basidiocarp development, spore formation, and geotropic dynamics of *P. cubensis* (Badham, 1982). Specifically, blue light and ultraviolet radiation (UV) have been found to facilitate fruiting body development (Badham, 1980), however, natural sources of UV might also lower the potency of psychoactive components (Rafati et al., 2009). Finally, spore prints (Fig. 5, B) can be collected (Fig. 3, D) from mature fruiting bodies (Fig. 5, C—H) for subsequent germination, or tissue samples can be taken from carpophores for cloning by culturing in liquid media or on semi-solid media (Fig. 3, D).

Protocols for different psychedelic mushroom species must be optimized for numerous stages of development, including spore germination, mycelial colonization, fruiting body initiation, and basidiocarp development. Further, understanding these factors will be critical to help increase uniformity and reproducibility for clinical research. Since

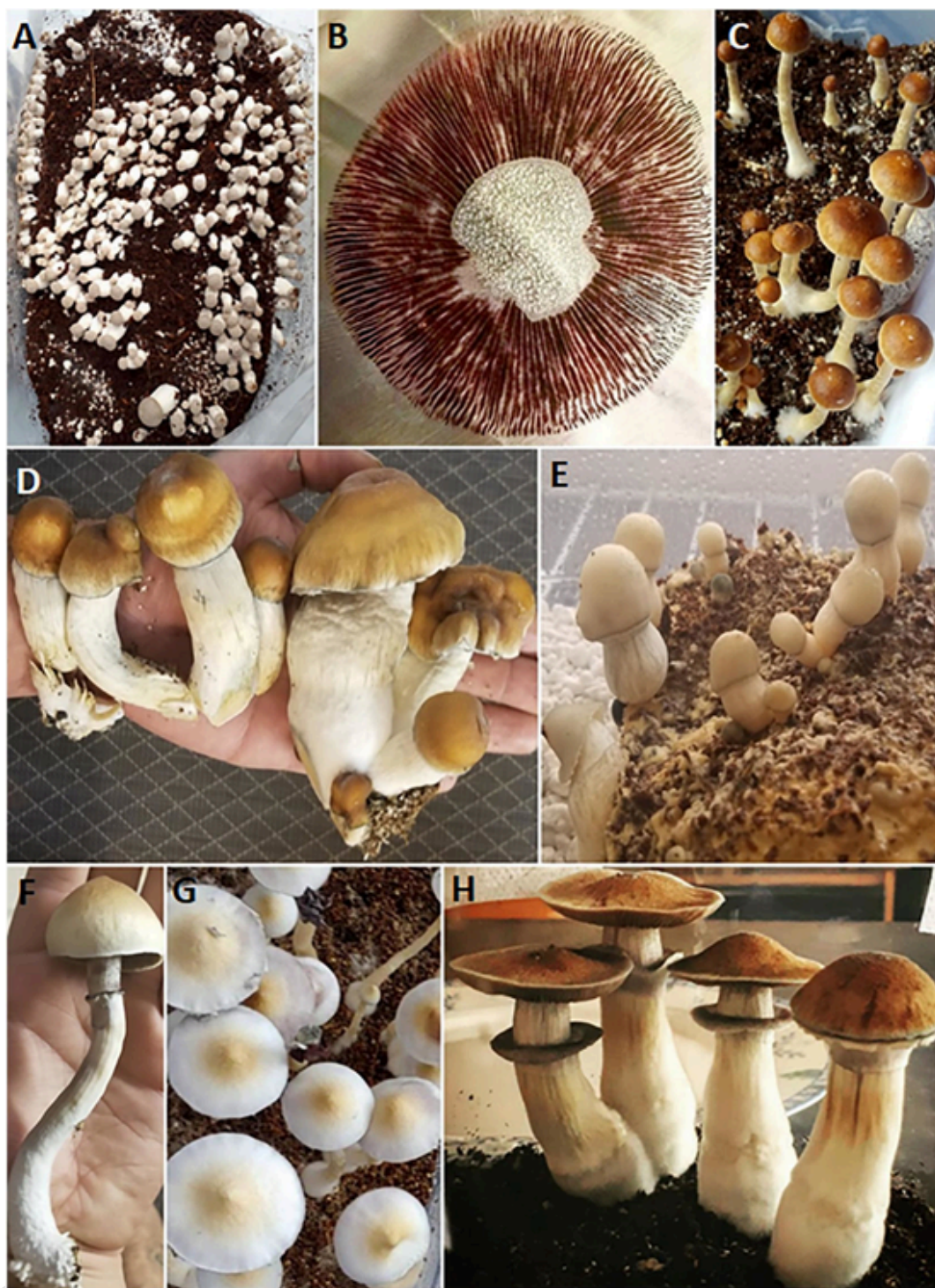


Fig. 5. Products of conventional cultivation, including different isolated lineages of *P. cubensis*. (A) Premature fruiting bodies developing from casing. (B) Spore print. (C) *P. cubensis* Golden Teacher. (D) *P. cubensis* Penis Envy. (E) *P. cubensis* Albino Penis Envy. (F) *P. cubensis* A+. (G) *P. cubensis* Albino A+. (H) *P. cubensis*.

mushroom morphogenesis is directed by environmental factors (Eger-Hummel, 1980), there is still much research that must be completed to determine the impact of specific environmental cues on growth and development of different species of psychedelic mushrooms. Microbial stimuli can impact development of *Psilocybe paneoliformis*, which can be replicated using activated charcoal or different light treatments (Eger-Hummel, 1980). Different types of fruiting substrates must be considered since yield and secondary metabolite profiles are dependent on lignocellulosic content in addition to macronutrients and trace mineral el-

ements (Sommano et al., 2022). Research relating to the optimization species-specific fruiting substrates for psychedelic mushrooms is lacking, and protocols must still be developed. By creating a completely standardized media solution and sequentially occluding individual and combinations of components, it is possible to refine media recipes and direct growth by manipulating substrates alone. Additionally, accumulation of volatile metabolites and high $[CO_2]$ can hamper development of fruiting bodies (Eger-Hummel, 1980; Jung and Son, 2021a, 2021b). For oyster mushrooms, maintaining $[CO_2]$ below 2100 ppm throughout

the fruiting phase enhances marketable value of the harvested products (Jung and Son, 2021a, 2021b). Additionally, temperature and humidity regulation are critical factors effecting yield of different mushrooms, optimization of which can double yield (Indra Setiawan, 2001). Light quality can also significantly influence basidiocarp development, with blue light being an important factor to consider (Badham, 1980; Eger-Hummel, 1980). Thus, abiotic conditions such as atmospheric factors and light quality must also be considered when developing optimized protocols for psychedelic mushroom production. Adapting a system presented by Pepe et al. (2022a); (Pepe et al., 2022b) to assess the responses of psychedelic mushrooms to abiotic factors can allow optimization of multiple abiotic factors for production of magic mushrooms.

In addition to optimizing protocols to enhance growth, development, and quality for different types of psychedelic mushrooms, breeding practices must also be developed. Compared to the short-lived monokaryotic state of Basidiomycetes, the dikaryotic state of fungi is a stable and long-living lifecycle phase in which binucleate cells contain nuclei of different, compatible mating types (Smith et al., 2002). Basidiomycetes can exhibit an array of different mating types which are governed by two unlinked loci, with one multiallelic locus controlling homeodomain transcription factor genes, and the other controlling a phosphorylation signal transduction cascade to activate pheromone responses (Kothe, 1996). *Psilocybe subaeruginosa* exhibits a tetrapolar mating type with a homeodomain locus containing *HD1* and *HD2*, and a pheromone/receptor locus with four *STE3* homologs (McTaggart et al., 2023). Certain autonomous *Psilocybin* species may not form hybrid dykaryons with one another (Gartz et al., 1994), although further work in this area is warranted. It is suggested that *Psilocybe* genetic improvement efforts will necessitate center of origin genetic species diversity, identification of specific genes related to traits of interest, and strategies to avoid inbreeding depression (McTaggart et al., 2023). It is possible that protoplast isolation and electroporation protocols and/or CRISPR/Cas systems can be used as genetic improvement strategies.

26. Chemical synthesis of psilocybin

Microclimate, agrochemicals, pests, disease, and variable growth conditions can lead to inconsistencies in psychoactive natural products (Scown and Keasling, 2022). Chemical synthesis of *Psilocybe* mushroom secondary metabolites currently represents the standard approach for deriving products for laboratory experiments and clinical trials. Chemical synthesis of psilocybin was first accomplished by Hofmann et al. (1959). In this classic paper, phosphorylation of psilocin to psilocybin, representing the most challenging step of synthesis, was accomplished with *O,O*-dibenzylphosphoryl chloride, which resulted in an unacceptable final yield of nearly 20% psilocybin (Nichols and Frescas, 1999). Nichols and Frescas (1999) improved this process by converting psilocin to 4-*O*-monobenzylphosphoryloxy-*N,N*-dimethyltryptamine, then to psilocybin by means of catalytic hydrolysis of the *O*-benzyl ester using palladium on carbon and methanol, to yield more than 45% psilocybin (Nichols and Frescas, 1999). Shiota et al. (2003) improved the former technique, allowing gram-scale synthesis of both psilocin and psilocybin. Psilocin was prepared by converting 4-hydroxyindole to 4-acetylindole using acetic anhydride, pyridine, and dichloromethane. Then, a two-step conversion from 4-acetylindole to 3-dimethylaminooxalyl-4-acetylindole using oxalyl chloride and *n*-hexane, followed by treatment with dimethylamine tetrahydrofuran solution was carried out. Treatment with lithium aluminum hydride and tetrahydrofuran yielded psilocin. Psilocin then yielded 85% psilocybin via intermediate *O,O*-dibenzyl phosphate and *N,O*-dibenzyl phosphate derivatives, representing key steps in psilocybin synthesis (Shiota et al., 2003). More recently, Kargbo et al. (2020) describe kilogram-scale synthesis of psilocybin to overcome challenges related to scaling up the previously described techniques under GMP conditions. The process in-

volves an optimized tree-step approach to synthesizing psilocin from 4-acetoxyindole, and direct phosphorylation of psilocin to psilocybin using a phosphoryl chloride and/or via hydrolysis in aqueous acidic environments. This process allowed psilocybin yields of over 99% (Kargbo et al., 2020). Although the chemical synthesis of psilocybin allows the production of GMP-compliant metabolites, these multi-step processes can be time consuming and laborious. Synthetic biology-mediated strategies for psilocybin production also offer alternative approaches for production of *Psilocybe* mushroom secondary metabolites.

27. Synthetic biology-mediated psilocybin production

If secondary metabolite isolation is impractical and/or a single psychoactive product represents the consumer's need, synthetic biology represents a powerful method for a robust supply chain (Jamieson et al., 2021). Moreover, recent advances in CRISPR/Cas-based methods (Nishida and Kondo, 2021) and computational strategies such as machine learning (Patra et al., 2023) have permitted exceptional progress related to synthetic biology for building effective and functional biological systems. An increasingly popular approach to synthetic biology-mediated engineering of bioproducts involves the Design-Build-Test-Learn (DBTL) strategy, which represents an effective, systematic tactic for designing biosystems (Cao et al., 2020). First, the biosynthesis pathway should be determined, which is a prerequisite for the heterologous production of targeted products. Fricke et al. (2017) elucidated the biosynthetic pathway of psilocybin through genome sequencing (previously outlined). However, there are potentially alternative routes for psilocybin production in different organisms such as the not yet elucidated psilocybin pathway in *Massospora* (Boyce et al., 2019). Discovering such pathways could help streamline production by means of synthetic biology. After the biosynthetic pathway is elucidated, an appropriate chassis (e.g., *Escherichia coli*, *Saccharomyces cerevisiae*, and *Aspergillus nidulans*) that meets the biosynthesis needs of interest must be selected (Jamieson et al., 2021). This is an important step, since different chassis organisms have positive and negative attributes that can either be conducive or unfavorable for bioproduction of specific compounds. Once a favorable chassis is chosen, the DBTL strategy can be enlisted (Jamieson et al., 2021) as a multidisciplinary framework to facilitate the development of efficient cell factories (Cao et al., 2020). This DBTL process must be iterated until sufficiently high production rates are achieved (Jamieson et al., 2021).

Methodologies enlisted throughout the DBTL cycle include computational design methods, genetic engineering, bioprocess testing, and omics analyses to help overcome specific production bottlenecks (Cao et al., 2020). CRISPR-based genome editing methods represent powerful tools enabling high precision genomic modification for strain construction during the "Build" phase (Jamieson et al., 2021). Screening approaches can then be enlisted during the "Test" phase to evaluate synthetic construct performance and quantify the obtained products (Jamieson et al., 2021). However, since the DBTL process must be sequentially repeated, continuous implementation may result in a surplus of information related to suboptimal construct designs instead of optimized breakthroughs (Cao et al., 2020). Machine learning (ML) can be enlisted to conquer such a treadmill by offering a platform for multi-scale modeling of process optimization by helping with analysis and conceptualization of large information databases (Cao et al., 2020; Gurdo et al., 2023; Jamieson et al., 2021). Such methods can be enlisted to facilitate biosynthesis of high-titer psychoactive substances using heterologous cell factories (Jamieson et al., 2021). In the following sections, recent advances in psilocybin production through heterologous microbial systems are reviewed. Then, the application of CRISPR-based methods and ML technologies are presented as promising approaches for producing high-titer psilocybin.

28. Psilocybin production in heterologous systems

To ensure a ready and consistent supply of tryptamines, various chemical synthesis and heterologous expression systems have been explored. Biocatalytic production of high-titer psilocybin (110 to 1.16×10^6 mg/L) has been achieved in heterologous microbial systems such as, *E. coli* (Adams et al., 2019), *S. cerevisiae* (Milne et al., 2020), and *A. nidulans* (Hoefgen et al., 2018). Strategies that have been developed for engineering heterologous expression of psilocybin are outlined in (Table 1).

Using *A. nidulans*, Hoefgen et al. (2018) established a method in which C-terminal 2A peptides are automatically removed after translation. A specifically designed split reporter gene in the vector also provides information about the correct polycistronic transcript translation, while allowing a reliable and fast screening process. With this system, a comparable titer of psilocybin to that produced in vivo by native *P. cubensis* was achieved (110 mg/L) psilocybin at 1.5% dry mycelial weight) (Hoefgen et al., 2018). Although the developed methodology by Hoefgen et al. (2018) streamlined the expression and assembly of multi-transgene, they did not evaluate the maximum number of genes that can be efficiently expressed by the developed system. Alternatively, Milne et al. (2020) reported titers of 580 ± 276 mg/L of psilocin and 627 ± 140 mg/L of psilocybin from 1 L scale fed-batch fermentation over nine days using *S. cerevisiae*. *P. cubensis*-derived psilocybin biosynthesis genes were expressed under strong promoters, and instead of the pathway decarboxylase PsiD expression, *Catharanthus roseus* -derived tryptophan decarboxylase (CrTDC) was expressed, which has high catalytic efficiency in yeast. Moreover, *P. cubensis*-derived cytochrome P450 reductase (CPR) was expressed to improve PsiH activity, while ARO1 and ARO2 were overexpressed to increase endogenous L-tryptophan levels. ARO1 and ARO2 overexpression combined with RIC1 (i.e., a shikimate pathway regulator) knockout through CRISPR-based tools can be considered an effective strategy for elevating L-tryptophan supply which results in increasing psilocybin production (Milne et al., 2020). Finally, Adams et al. (2019) combined two powerful strategies (i.e., metabolic engineering and heterologous expression) which resulted in 1.16×10^6 mg/L psilocybin in *E. coli* over several days from gradually applied 4-hydroxyindole in a 1.5 L bioreactor (Adams et al., 2019). This method used an endogenous bacterial enzyme, TrpB, to convert 4-hydroxyindole into 4-hydroxy-L-tryptophan.

Table 1
Application of microbial heterologous systems in *Psilocybe* sp.

Microbial host	Advantages	Disadvantage	Required steps	References
<i>Aspergillus nidulans</i>	• Fermentable, • Correct splicing of fungal introns, • Robust secondary metabolism	• Production of aflatoxin precursors, • Variability in morphology may affect product titer	10	Hoefgen et al. (2018)
<i>Saccharomyces cerevisiae</i>	• Functional expression of P450s, • Myriad synthetic biology tools, • History of use in food biotechnology	• Unable to splice introns, • Low initial titers in lab strains, • Toxicity limits	20	Milne et al. (2020)
<i>Escherichia coli</i>	• Simple and predictable organism, • Myriad synthetic biology tools, • High growth rate	• Unable to splice introns, • Unable to functionally express many eukaryotic enzymes	5	Adams et al. (2019)

Conversion of norbaeocystin to psilocybin was facilitated by PsiM, PsiK, and PsiD, which were heterologously expressed on a high copy plasmid containing a single T7 promoter. To further facilitate high-titer production, methionine and serine were exogenously supplemented into the media, which can be anabolized into SAM by MetK, a native *E. coli* enzyme, while the by-product, S-adenosylhomocysteine (SAH), can further be recycled into more methionine by *E. coli* enzymes MetE, LuxS, and Mtn. Ultimately, this method allowed high titers of psilocybin to be produced using *E. coli* (Adams et al., 2019). Although *E. coli* is a simple and predictable organism which makes it an appropriate chassis for high-titer psilocybin production, problems in splicing introns and functionally may arise due to expression of various eukaryotic enzymes. These factors could present challenges for genetically engineering such pathways into this microbial host.

Generally, one of the main challenges in synthetic biology through microbial heterologous systems is the need for efficient production of biomolecules such as proteins, enzymes, and metabolites (Mahdavi et al., 2022). Microbial heterologous systems rely on living cells, which can be time-consuming, expensive, and difficult to scale up (Lambertz et al., 2014). Cell-free systems offer a solution to these challenges by allowing the production of secondary metabolites and other biomolecules in vitro without the need for living cells (Laohakunakorn et al., 2020; Lu, 2017; Tinafar et al., 2019). Although there is no report that uses cell-free systems for psilocybin production, this approach can offer several advantages, including increased efficiency, scalability, and flexibility. In addition, cell-free systems can be designed to optimize psilocybin production by modifying the enzymes and pathways involved in synthesizing the compound. These advantages make cell-free systems an attractive platform for the production of psilocybin and other bioactive compounds, paving the way for their use in therapeutic applications.

29. CRISPR-based genome editing as a promising approach for the metabolic engineering of psilocybin

Various CRISPR-based genome editing methods (Fig. 6) can be considered rapid and precise approaches in metabolic engineering of psilocybin. Although different CRISPR-based methods have been successfully employed for genome editing in filamentous fungi such as *Neurospora crassa* (Matsu-ura et al., 2015), *Trichoderma reesei* (Liu et al., 2015), and *Aspergillus nidulans* (Katayama et al., 2016; Nødvig et al., 2015), there exists no report that uses CRISPR/Cas methods to produce psilocybin. Several excellent review articles present different applications of CRISPR/Cas-based methods for metabolic engineering and synthetic biology (Chu, 2022; Liu et al., 2022; Nishida and Kondo, 2021). The methodologies presented in these review papers can also be employed for metabolic engineering-mediated psilocybin production. For instance, CRISPR-mediated knockout of Psi (e.g., PsiD, PsiH, PsiM) genes would confer reduced production or complete loss of psilocybin and one or more of its related tryptamines in *Psilocybe* mushrooms. The availability of these engineered mushrooms could be of benefit to the neuroscience research community in order to better understand the impact of psilocybin and psilocybin-free mushroom extracts on brain-related molecular and biochemical processes as a strategy to develop safer and more precise therapeutic applications. Moreover, it may also reduce the potential for substance abuse of psilocybin-containing mushrooms, as PsiM mutant strains would yield non-psychoactive mushrooms while preserving other secondary metabolites. On the other hand, CRISPR can be used to overexpress the psi genes to increase the psilocybin production in the edited mushrooms. Such methodologies are promising for the development of tailored *Psilocybe* strains with specific metabolic traits that could pave the way for new discoveries in the field of psychedelic medicine.

Although previous studies have utilized the conventional homologous recombination-based gene disruption approach to analyze functional genes in certain mushrooms such as *Coprinopsis cinerea*

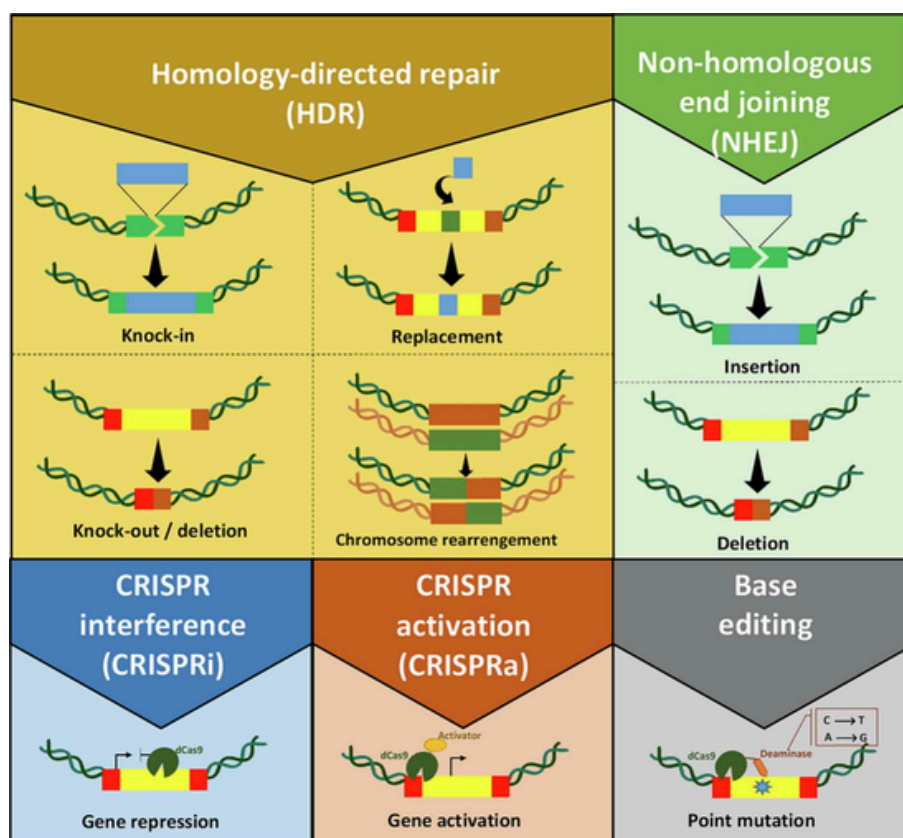


Fig. 6. Different types of CRISPR-mediated metabolic engineering. Random indels occur at the cleavage site through non-homologous end joining (NHEJ) when double-strand break (DSB) is induced. Precision editing is achievable with homology-directed repair (HDR) by substituting homologous templates, which enables diverse modes of genome editing like inter-chromosomal recombination, replacement, complete deletion, and marker-less knock-in. In some organisms, nickases can also effectively stimulate HDR. Additionally, a pair of nickases aimed at nearby and opposing sites can cause DSB. DNA deamination is utilized in base editing to introduce precise point mutations. Innate gene regulation can be overridden via CRISPR activation (CRISPRa) and CRISPR interference (CRISPRi) by obstructing transcription or bringing in transcription machinery at targeted regions. Parts of the scheme were generated using [BioRender.com](#).

(Nakazawa et al., 2011) and *Schizophyllum commune* (Ohm et al., 2010), its application to other mushroom species has proven to be challenging (Wang et al., 2020). However, with the emergence of CRISPR base editing, the latest generation genome editing tool, has shown great promise in editing various organisms, including plants, animals and bacteria (Hesami et al., 2023; Wang et al., 2020). Despite this, there have only been a few reports of CRISPR base editing of functional genes in mushrooms. For instance, recent studies highlighted the successful disruption of *ppo*, which encodes polyphenol oxidase in *Agaricus bisporus* (Hall, 2016; Schuster and Kahmann, 2019; Waltz, 2016), but no research papers or patents have been published on this topic. Furthermore, a study on *S. commune* demonstrated that preassembled Cas9 ribonucleoproteins (RNPs) were able to disrupt the *hom2* gene, a homeodomain transcription factor, but the editing efficiency was low and the preparation of RNPs was a time-consuming and expensive process (Jan Vonk et al., 2019). These findings highlight the challenges of conducting functional gene studies in mushrooms. Therefore, there is a pressing need to establish an efficient functional gene editing technology in mushrooms, especially in *Psilocybe* sp., to promote biological research and biotechnology development.

30. Machine learning-assisted metabolic engineering for psilocybin production

Systems metabolic engineering, which facilitates the development of strains through the integration of evolutionary engineering, synthetic biology, and system biology are among the most significant methods of biorefinery establishment approaches (Jang et al., 2022). As

with other biological disciplines, the availability of bio big data has also exponentially increased in the field of systems metabolic engineering (Kim et al., 2020). Bio big data consist of (but is not limited to) different multi-omics datasets that can be employed for the application of ML methods (Hesami et al., 2022). A wide range of ML methods has streamlined the exploration of large metabolic space and optimization of bio-process parameters that are linked to metabolic pathways of the bio-products of interest (Kim et al., 2020). Although previous studies have successfully employed ML algorithms (e.g., support vector machine, artificial neural networks) for different metabolic engineering purposes such as fermentation (Li et al., 2019; Pappu and Gummadi, 2017; Zheng et al., 2017), metabolic flux optimization (Costello and Martin, 2018; Groher et al., 2019; Jervis et al., 2019a; Jervis et al., 2019b; Zhou et al., 2018), metabolic pathway reconstruction (Mellor et al., 2016; Segler et al., 2018; Wu et al., 2019), gene annotation (Blin et al., 2019; Skinnider et al., 2017), and host strain selection (Clauwaert et al., 2019; Ryu et al., 2019) (Fig. 7), there are no studies that use ML in metabolic engineering for psilocybin production. Metabolic engineering of psilocybin can similarly be enhanced by employing ML methods. However, it is essential to generate high-quality datasets to properly apply ML in metabolic engineering of psilocybin and other mushroom secondary metabolites. Additionally, the downstream bioprocesses (e.g., purification and product recovery) that require trial and error to achieve optimization can be improved using ML methods.

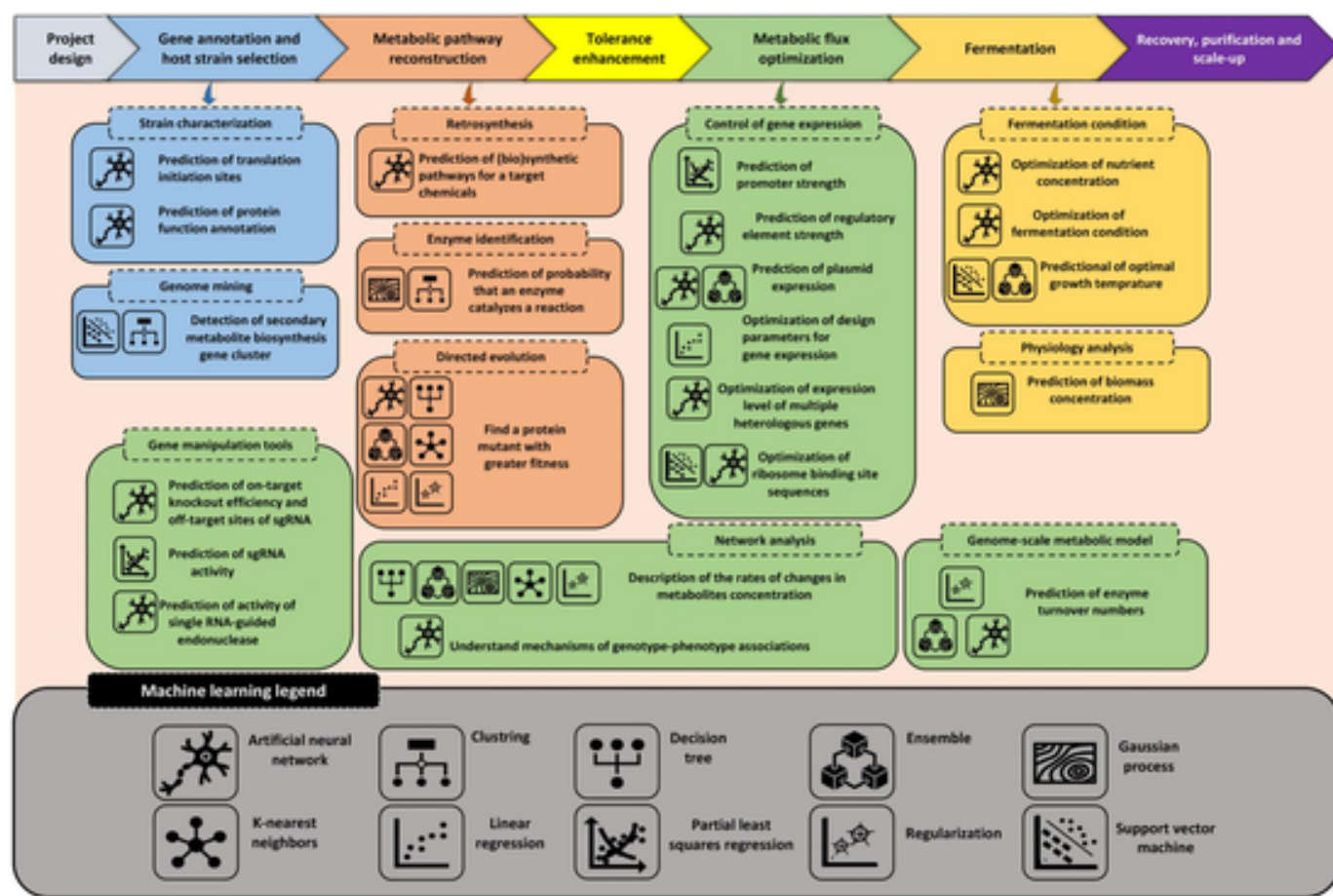


Fig. 7. Application of various machine learning (ML) algorithms in different steps of metabolic engineering including gene annotation and host strain selection, metabolic pathway reconstruction, metabolic flux optimization, and fermentation. ML tools used in each step of metabolic engineering are presented with relevant symbols. The scheme was adapted from Kim et al. (2020).

31. Conclusion

Psychedelic mushrooms have recently re-emerged as a high interest area of research to a broad spectrum of disciplines, including the fields of mycology, biotechnology, biochemistry, biomedical, neuroscience, and psychology. In order to improve and adapt the utility of psychedelic mushrooms to specific applications, there are certain knowledge gaps that must be filled by perpetuating research efforts. Although there has been considerable progress in research related to human health, biochemistry, and biotechnology, there are certain bottlenecks in progress that must be overcome. Machine learning and synthetic biology represent the keys to unlock most aspects of psychedelic mushroom research. Using machine learning and synthetic biology as respective tools for pathway reconstruction and bioengineering those precise pathways through heterologous systems, unique opportunities to study different molecular mechanisms of metabolite biosynthesis arise. This avenue lends itself to developing refined strategies for the production of specific compounds of interest with high purity standards that would become more widely available for use in pre-clinical and clinical studies. The availability of specific, highly purified secondary metabolites from accredited sources would enhance the replicability of these studies while offering the potential to add additional layers to the research by including or excluding precise compounds, to better identify their individual effects or potential entourage effects. Higher quality and volume of research relating to different applications of psychedelic mushrooms will undoubtedly generate greater interest and more acceptance toward psychedelics as important medicines that can change the world.

Author credits

Marco Pepe and Mohsen Hesami conceptualized the work. Andrew Maxwell Phineas Jones supervised and mentored the work. Marco Pepe, Mohsen Hesami, and Karla A. de la Cerda wrote the original manuscript. Melissa L. Perreault, Andrew Maxwell Phineas Jones, and Tom Hsiang reviewed the manuscript. Andrew Maxwell Phineas Jones, Marco Pepe, and Mohsen Hesami edited the manuscript. All authors contributed to the manuscript and approved the final version.

Declaration of Competing Interest

The authors declare that we have no conflict of interest.

Data availability

All data is presented in the published work.

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